Material Characterization Report

NanoVNA UTN Toolkit

2026-10-24 12:11:08

Measurement Details:

- **Sample:** Muestra4
- **Technique:** Open-ended coaxial probe (liquids)
- **Temperature:** 23.0 °C
- **Reference Liquids:** Distilled Water / Isopropyl Alcohol (IPA)
- **Probe:** 3 mm probe
- **Frequency Range:** 0.005–6.000 GHz, 1200 pts
- **Data sources:**
  - **Open:** kit:cal_sonda3mm_UBA_1200p
  - **Short:** kit:cal_sonda3mm_UBA_1200p (1 spike(s) replaced in the computation: 4885 MHz)
  - **Ref 1:** kit:cal_sonda3mm_UBA_1200p
  - **Ref 2:** kit:cal_sonda3mm_UBA_1200p
  - **DUT:** imported

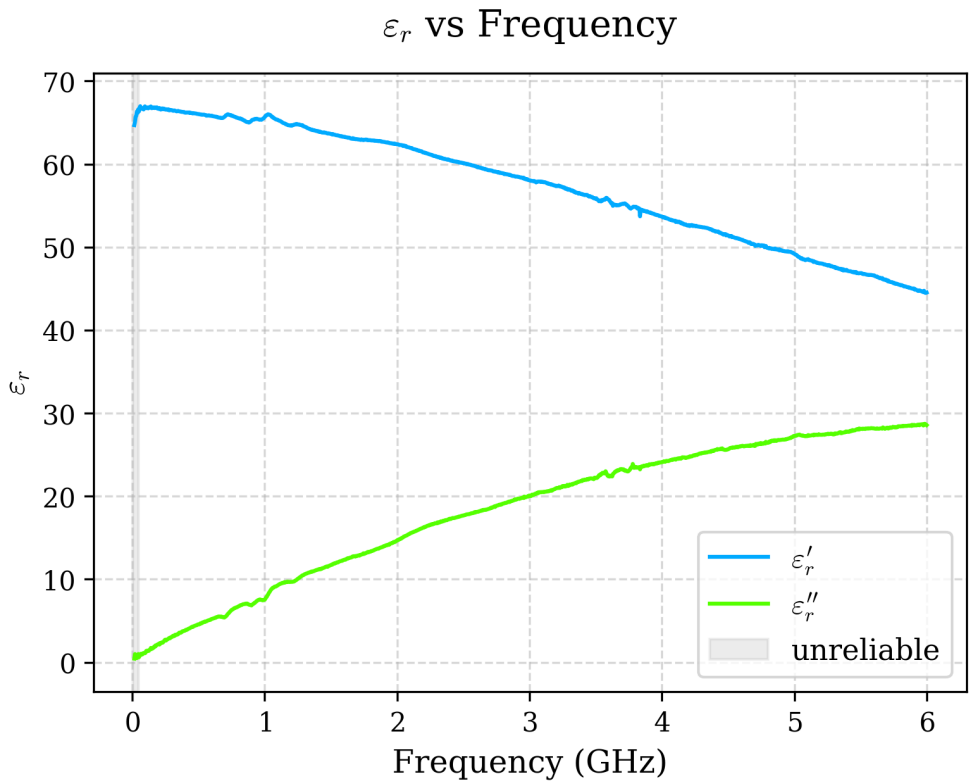
Contents

1 Measurement Results **2**
1.1 Complex Permittivity $\varepsilon_r(f)$ 2
1.2 S_{11} — Smith Chart 29

2 Measurement Quality **56**

1 Measurement Results

1.1 Complex Permittivity $\varepsilon_r(f)$



Points without a solution: 2 of 1200 (left blank in the plot and shown as — in the table).

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.005	—	—	—
0.010	—	—	—
0.015	64.6915	0.4360	0.0067
0.020	65.0737	0.9848	0.0151
0.025	65.6846	0.4712	0.0072
0.030	65.8332	0.6914	0.0105
0.035	66.3396	0.5136	0.0077
0.040	66.1738	0.8062	0.0122
0.045	66.5997	0.9887	0.0148
0.050	66.3460	0.6008	0.0091
0.055	66.8050	0.7802	0.0117
0.060	66.9724	0.8346	0.0125
0.065	66.6985	1.0243	0.0154
0.070	66.6554	0.9485	0.0142
0.075	66.7313	0.9357	0.0140
0.080	66.7397	1.0558	0.0158
0.085	66.5795	1.1211	0.0168

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.090	66.7714	1.0969	0.0164
0.095	66.9522	1.0875	0.0162
0.100	66.8033	1.1468	0.0172
0.105	66.7377	1.3671	0.0205
0.110	66.7553	1.3806	0.0207
0.115	66.8251	1.3670	0.0205
0.120	66.7101	1.4050	0.0211
0.125	66.8411	1.3758	0.0206
0.130	66.8302	1.4903	0.0223
0.135	66.8702	1.5519	0.0232
0.140	66.9426	1.7233	0.0257
0.145	66.7326	1.5910	0.0238
0.150	66.7459	1.6545	0.0248
0.155	66.8093	1.6730	0.0250
0.160	66.7597	1.8030	0.0270
0.165	66.8327	1.8802	0.0281
0.170	66.7453	1.8437	0.0276
0.175	66.7929	2.0271	0.0303
0.180	66.7909	1.9927	0.0298
0.185	66.8371	2.0811	0.0311
0.190	66.6868	2.1416	0.0321
0.195	66.6893	2.1742	0.0326
0.200	66.6418	2.2312	0.0335
0.205	66.7540	2.1274	0.0319
0.210	66.6739	2.3110	0.0347
0.215	66.5663	2.2807	0.0343
0.220	66.6279	2.3741	0.0356
0.225	66.7074	2.3820	0.0357
0.230	66.6206	2.3619	0.0355
0.235	66.6883	2.4611	0.0369
0.240	66.6351	2.5570	0.0384
0.245	66.6376	2.6787	0.0402
0.250	66.5612	2.7264	0.0410
0.255	66.6037	2.6159	0.0393
0.260	66.5854	2.6977	0.0405
0.265	66.5811	2.7507	0.0413
0.270	66.5151	2.8167	0.0423
0.275	66.5840	2.8643	0.0430
0.280	66.4711	2.9326	0.0441
0.285	66.5299	2.8885	0.0434
0.290	66.5169	3.0141	0.0453
0.295	66.5083	3.0102	0.0453
0.300	66.4882	3.0580	0.0460
0.305	66.4416	3.1578	0.0475
0.310	66.4193	3.1988	0.0482
0.315	66.4584	3.1432	0.0473

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.320	66.4467	3.1887	0.0480
0.325	66.4210	3.3159	0.0499
0.330	66.3811	3.3059	0.0498
0.335	66.3514	3.3065	0.0498
0.340	66.3677	3.3648	0.0507
0.345	66.4065	3.3665	0.0507
0.350	66.3773	3.4786	0.0524
0.355	66.3462	3.5699	0.0538
0.360	66.3399	3.5420	0.0534
0.365	66.3333	3.5471	0.0535
0.370	66.2888	3.5880	0.0541
0.375	66.2979	3.6987	0.0558
0.380	66.2801	3.7338	0.0563
0.385	66.2747	3.7439	0.0565
0.390	66.2758	3.7822	0.0571
0.395	66.2160	3.7887	0.0572
0.400	66.1935	3.8487	0.0581
0.405	66.1964	3.9085	0.0590
0.410	66.2173	3.8951	0.0588
0.415	66.2040	3.9296	0.0594
0.420	66.2160	4.0000	0.0604
0.425	66.2062	4.0418	0.0610
0.430	66.2184	4.0748	0.0615
0.435	66.1484	4.1042	0.0620
0.440	66.1727	4.1615	0.0629
0.445	66.1564	4.2272	0.0639
0.450	66.1447	4.2661	0.0645
0.455	66.1018	4.2361	0.0641
0.460	66.1375	4.2890	0.0649
0.465	66.1232	4.3455	0.0657
0.470	66.1037	4.3807	0.0663
0.475	66.0988	4.4106	0.0667
0.480	66.0927	4.4417	0.0672
0.485	66.0682	4.4506	0.0674
0.490	66.0442	4.4997	0.0681
0.495	66.0391	4.5538	0.0690
0.500	66.0168	4.5454	0.0689
0.505	66.0222	4.5691	0.0692
0.510	65.9909	4.6111	0.0699
0.515	65.9766	4.6965	0.0712
0.520	65.9796	4.6715	0.0708
0.525	65.9434	4.7065	0.0714
0.530	65.9468	4.7945	0.0727
0.535	65.9474	4.8086	0.0729
0.540	65.9559	4.8268	0.0732
0.545	65.9394	4.8841	0.0741

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.550	65.9111	4.8680	0.0739
0.555	65.8994	4.9149	0.0746
0.560	65.8288	4.9318	0.0749
0.565	65.8468	4.9821	0.0757
0.570	65.8395	5.0228	0.0763
0.575	65.8688	5.0487	0.0766
0.580	65.8663	5.0799	0.0771
0.585	65.8236	5.0927	0.0774
0.590	65.8169	5.1079	0.0776
0.595	65.8274	5.1794	0.0787
0.600	65.8295	5.2143	0.0792
0.605	65.7838	5.2352	0.0796
0.610	65.7690	5.2645	0.0800
0.615	65.7552	5.2943	0.0805
0.620	65.7362	5.3133	0.0808
0.625	65.7202	5.3523	0.0814
0.630	65.7206	5.3621	0.0816
0.635	65.6809	5.4324	0.0827
0.640	65.6925	5.4471	0.0829
0.645	65.6267	5.4730	0.0834
0.650	65.6156	5.5098	0.0840
0.655	65.5943	5.5131	0.0840
0.660	65.5913	5.4865	0.0836
0.665	65.5903	5.4999	0.0839
0.670	65.5448	5.5044	0.0840
0.675	65.5528	5.4533	0.0832
0.680	65.5458	5.4683	0.0834
0.685	65.5707	5.4135	0.0826
0.690	65.6218	5.4054	0.0824
0.695	65.6817	5.3868	0.0820
0.700	65.7429	5.4507	0.0829
0.705	65.8499	5.5250	0.0839
0.710	65.8950	5.5678	0.0845
0.715	65.9457	5.6825	0.0862
0.720	65.9980	5.7802	0.0876
0.725	65.9823	5.8804	0.0891
0.730	65.9503	5.9462	0.0902
0.735	65.9089	6.0502	0.0918
0.740	65.9100	6.1333	0.0931
0.745	65.9006	6.2265	0.0945
0.750	65.8177	6.2976	0.0957
0.755	65.8138	6.3383	0.0963
0.760	65.8061	6.3863	0.0970
0.765	65.7499	6.4499	0.0981
0.770	65.7095	6.5099	0.0991
0.775	65.6952	6.5179	0.0992

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.780	65.6679	6.5791	0.1002
0.785	65.6025	6.6013	0.1006
0.790	65.5953	6.6238	0.1010
0.795	65.5637	6.6691	0.1017
0.800	65.5321	6.7070	0.1023
0.805	65.5016	6.7593	0.1032
0.810	65.4884	6.7898	0.1037
0.815	65.5091	6.8557	0.1047
0.820	65.4503	6.8797	0.1051
0.825	65.3936	6.8861	0.1053
0.830	65.4067	6.9381	0.1061
0.835	65.3247	6.9811	0.1069
0.840	65.2835	6.9998	0.1072
0.845	65.2682	7.0018	0.1073
0.850	65.1933	7.0268	0.1078
0.855	65.1644	7.0619	0.1084
0.860	65.0905	7.0266	0.1080
0.865	65.0725	7.0690	0.1086
0.870	65.0650	7.0559	0.1084
0.875	65.0533	6.9802	0.1073
0.880	65.0051	6.9442	0.1068
0.885	65.0320	6.9126	0.1063
0.890	65.0908	6.9159	0.1062
0.895	65.1382	6.8750	0.1055
0.900	65.2354	6.8452	0.1049
0.905	65.3190	6.9240	0.1060
0.910	65.3443	6.9870	0.1069
0.915	65.3559	7.0622	0.1081
0.920	65.3619	7.0836	0.1084
0.925	65.4376	7.1797	0.1097
0.930	65.4525	7.2181	0.1103
0.935	65.4334	7.2951	0.1115
0.940	65.4644	7.3797	0.1127
0.945	65.4266	7.4258	0.1135
0.950	65.4108	7.4642	0.1141
0.955	65.3890	7.5240	0.1151
0.960	65.3417	7.5140	0.1150
0.965	65.3556	7.5663	0.1158
0.970	65.3533	7.5301	0.1152
0.975	65.3375	7.5084	0.1149
0.980	65.3972	7.4922	0.1146
0.985	65.4486	7.4588	0.1140
0.990	65.4906	7.4999	0.1145
0.995	65.5763	7.5057	0.1145
1.000	65.6648	7.5309	0.1147
1.005	65.7695	7.6543	0.1164

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.010	65.8661	7.7832	0.1182
1.015	65.8885	7.8768	0.1195
1.020	65.9519	8.0316	0.1218
1.025	66.0011	8.1453	0.1234
1.030	65.9787	8.3092	0.1259
1.035	65.9633	8.4188	0.1276
1.040	65.9262	8.5577	0.1298
1.045	65.8543	8.6724	0.1317
1.050	65.7909	8.8046	0.1338
1.055	65.6763	8.8588	0.1349
1.060	65.6685	8.9631	0.1365
1.065	65.5721	8.9699	0.1368
1.070	65.5473	9.0375	0.1379
1.075	65.4735	9.1079	0.1391
1.080	65.4302	9.1134	0.1393
1.085	65.3744	9.1293	0.1396
1.090	65.3793	9.1803	0.1404
1.095	65.3385	9.2403	0.1414
1.100	65.3192	9.2893	0.1422
1.105	65.2588	9.3333	0.1430
1.110	65.2195	9.3891	0.1440
1.115	65.2045	9.4384	0.1448
1.120	65.1649	9.4553	0.1451
1.125	65.1137	9.4853	0.1457
1.130	65.0475	9.5405	0.1467
1.135	64.9952	9.5701	0.1472
1.140	64.9489	9.5489	0.1470
1.145	64.9216	9.6208	0.1482
1.150	64.8870	9.5901	0.1478
1.155	64.8696	9.6541	0.1488
1.160	64.7898	9.6293	0.1486
1.165	64.7801	9.6829	0.1495
1.170	64.7437	9.6530	0.1491
1.175	64.6853	9.6898	0.1498
1.180	64.6795	9.6953	0.1499
1.185	64.6717	9.6657	0.1495
1.190	64.6414	9.6706	0.1496
1.195	64.6516	9.6770	0.1497
1.200	64.6901	9.6795	0.1496
1.205	64.6220	9.6806	0.1498
1.210	64.6840	9.6905	0.1498
1.215	64.6973	9.7139	0.1501
1.220	64.7475	9.7183	0.1501
1.225	64.7523	9.7619	0.1508
1.230	64.7894	9.8351	0.1518
1.235	64.7676	9.8629	0.1523

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.240	64.8303	9.9190	0.1530
1.245	64.8255	9.9811	0.1540
1.250	64.7707	10.0561	0.1553
1.255	64.7664	10.1131	0.1561
1.260	64.7743	10.1470	0.1567
1.265	64.7280	10.2348	0.1581
1.270	64.7138	10.2828	0.1589
1.275	64.7000	10.3472	0.1599
1.280	64.6692	10.4065	0.1609
1.285	64.6866	10.4661	0.1618
1.290	64.5776	10.4731	0.1622
1.295	64.5426	10.5332	0.1632
1.300	64.5268	10.5830	0.1640
1.305	64.4837	10.6182	0.1647
1.310	64.4644	10.6601	0.1654
1.315	64.4225	10.6898	0.1659
1.320	64.3443	10.7099	0.1664
1.325	64.3244	10.7503	0.1671
1.330	64.2913	10.7871	0.1678
1.335	64.2538	10.8009	0.1681
1.340	64.2366	10.8190	0.1684
1.345	64.1779	10.8705	0.1694
1.350	64.1577	10.9053	0.1700
1.355	64.1165	10.8839	0.1698
1.360	64.1344	10.9230	0.1703
1.365	64.0915	10.9606	0.1710
1.370	64.0658	10.9670	0.1712
1.375	64.0771	10.9994	0.1717
1.380	64.0392	11.0362	0.1723
1.385	64.0167	11.0795	0.1731
1.390	63.9798	11.0940	0.1734
1.395	63.9816	11.1309	0.1740
1.400	63.9861	11.1434	0.1742
1.405	63.9196	11.1838	0.1750
1.410	63.9301	11.2153	0.1754
1.415	63.8651	11.1962	0.1753
1.420	63.8929	11.2341	0.1758
1.425	63.8559	11.2204	0.1757
1.430	63.8222	11.2996	0.1770
1.435	63.7943	11.3000	0.1771
1.440	63.8041	11.3435	0.1778
1.445	63.7891	11.3681	0.1782
1.450	63.8221	11.4036	0.1787
1.455	63.7702	11.4412	0.1794
1.460	63.7743	11.4917	0.1802
1.465	63.7531	11.5185	0.1807

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.470	63.7133	11.5214	0.1808
1.475	63.6862	11.4913	0.1804
1.480	63.7108	11.5797	0.1818
1.485	63.6797	11.6070	0.1823
1.490	63.6759	11.6330	0.1827
1.495	63.6480	11.6708	0.1834
1.500	63.6287	11.7155	0.1841
1.505	63.6134	11.7462	0.1847
1.510	63.5615	11.7515	0.1849
1.515	63.6115	11.8336	0.1860
1.520	63.5596	11.8670	0.1867
1.525	63.5490	11.9101	0.1874
1.530	63.5309	11.8910	0.1872
1.535	63.5059	11.9355	0.1879
1.540	63.4742	11.9633	0.1885
1.545	63.4677	12.0247	0.1895
1.550	63.4750	12.0203	0.1894
1.555	63.4400	12.0407	0.1898
1.560	63.4231	12.0989	0.1908
1.565	63.3922	12.1421	0.1915
1.570	63.3613	12.1653	0.1920
1.575	63.3634	12.1895	0.1924
1.580	63.3680	12.2074	0.1926
1.585	63.3083	12.2177	0.1930
1.590	63.2983	12.2833	0.1941
1.595	63.2973	12.3033	0.1944
1.600	63.2768	12.3411	0.1950
1.605	63.2826	12.3511	0.1952
1.610	63.2373	12.3519	0.1953
1.615	63.2290	12.4129	0.1963
1.620	63.2260	12.4770	0.1973
1.625	63.2059	12.5150	0.1980
1.630	63.1620	12.4903	0.1977
1.635	63.1860	12.5478	0.1986
1.640	63.1814	12.5664	0.1989
1.645	63.1316	12.5944	0.1995
1.650	63.0764	12.6077	0.1999
1.655	63.0662	12.6284	0.2002
1.660	63.0445	12.6939	0.2013
1.665	63.1247	12.7543	0.2020
1.670	63.0674	12.7437	0.2021
1.675	63.0218	12.7705	0.2026
1.680	63.0455	12.7739	0.2026
1.685	63.0004	12.7286	0.2020
1.690	63.0055	12.8052	0.2032
1.695	63.0108	12.7943	0.2030

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.700	62.9876	12.8155	0.2035
1.705	62.9630	12.8564	0.2042
1.710	62.9778	12.8615	0.2042
1.715	62.9539	12.8385	0.2039
1.720	62.9311	12.9200	0.2053
1.725	62.9071	12.9323	0.2056
1.730	62.9755	12.9691	0.2059
1.735	62.9313	12.9921	0.2064
1.740	62.9156	13.0365	0.2072
1.745	62.8852	13.1145	0.2085
1.750	62.9107	13.0427	0.2073
1.755	62.9407	13.1094	0.2083
1.760	62.9285	13.1166	0.2084
1.765	62.9613	13.1509	0.2089
1.770	62.9058	13.1875	0.2096
1.775	62.9338	13.2579	0.2107
1.780	62.8956	13.2470	0.2106
1.785	62.8742	13.2944	0.2114
1.790	62.8756	13.2810	0.2112
1.795	62.8845	13.3230	0.2119
1.800	62.8643	13.3251	0.2120
1.805	62.8501	13.3886	0.2130
1.810	62.8499	13.3996	0.2132
1.815	62.8539	13.4256	0.2136
1.820	62.8444	13.4572	0.2141
1.825	62.8430	13.4919	0.2147
1.830	62.8027	13.5428	0.2156
1.835	62.8118	13.5716	0.2161
1.840	62.7996	13.6043	0.2166
1.845	62.8007	13.6435	0.2173
1.850	62.7907	13.6637	0.2176
1.855	62.7912	13.6843	0.2179
1.860	62.7632	13.6795	0.2180
1.865	62.7872	13.7387	0.2188
1.870	62.7817	13.7756	0.2194
1.875	62.7459	13.8123	0.2201
1.880	62.7473	13.8264	0.2203
1.885	62.6919	13.8739	0.2213
1.890	62.6967	13.8794	0.2214
1.895	62.6794	13.9261	0.2222
1.900	62.6762	13.9947	0.2233
1.905	62.6648	13.9824	0.2231
1.910	62.6382	14.0243	0.2239
1.915	62.6153	14.0822	0.2249
1.920	62.6084	14.0776	0.2249
1.925	62.6239	14.1223	0.2255

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.930	62.5865	14.1218	0.2256
1.935	62.5929	14.1824	0.2266
1.940	62.5385	14.1452	0.2262
1.945	62.5290	14.2648	0.2281
1.950	62.5089	14.2863	0.2285
1.955	62.5123	14.3278	0.2292
1.960	62.5044	14.3328	0.2293
1.965	62.4594	14.3952	0.2305
1.970	62.4679	14.4002	0.2305
1.975	62.4724	14.4930	0.2320
1.980	62.4186	14.4823	0.2320
1.985	62.4499	14.4942	0.2321
1.990	62.4249	14.5635	0.2333
1.995	62.4005	14.6194	0.2343
2.000	62.3724	14.6873	0.2355
2.005	62.3800	14.6840	0.2354
2.010	62.3713	14.7766	0.2369
2.015	62.3109	14.7703	0.2370
2.020	62.2926	14.8239	0.2380
2.025	62.3017	14.8770	0.2388
2.030	62.2989	14.8822	0.2389
2.035	62.2857	14.9705	0.2404
2.040	62.2556	14.9895	0.2408
2.045	62.1765	15.0464	0.2420
2.050	62.2441	15.1334	0.2431
2.055	62.2153	15.1493	0.2435
2.060	62.1517	15.1947	0.2445
2.065	62.1933	15.2281	0.2449
2.070	62.1002	15.2611	0.2458
2.075	62.0476	15.2945	0.2465
2.080	62.0485	15.3495	0.2474
2.085	62.0329	15.4078	0.2484
2.090	62.0086	15.4336	0.2489
2.095	61.9846	15.4722	0.2496
2.100	61.9507	15.5454	0.2509
2.105	61.9293	15.5714	0.2514
2.110	61.9389	15.5964	0.2518
2.115	61.8691	15.6428	0.2528
2.120	61.8168	15.6537	0.2532
2.125	61.8023	15.7040	0.2541
2.130	61.7716	15.7306	0.2547
2.135	61.7750	15.7661	0.2552
2.140	61.7586	15.7928	0.2557
2.145	61.7067	15.8453	0.2568
2.150	61.7154	15.8600	0.2570
2.155	61.6588	15.9082	0.2580

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.160	61.6239	15.9460	0.2588
2.165	61.6280	15.9829	0.2593
2.170	61.5734	16.0193	0.2602
2.175	61.5734	16.0701	0.2610
2.180	61.5458	16.0921	0.2615
2.185	61.5422	16.1163	0.2619
2.190	61.4819	16.1635	0.2629
2.195	61.4070	16.2050	0.2639
2.200	61.4096	16.2445	0.2645
2.205	61.3759	16.2308	0.2645
2.210	61.3811	16.3124	0.2658
2.215	61.3221	16.3202	0.2661
2.220	61.3251	16.3522	0.2666
2.225	61.2887	16.4097	0.2677
2.230	61.2554	16.4423	0.2684
2.235	61.2364	16.4584	0.2688
2.240	61.2020	16.4594	0.2689
2.245	61.1687	16.5330	0.2703
2.250	61.1538	16.5399	0.2705
2.255	61.1363	16.5482	0.2707
2.260	61.0949	16.6217	0.2721
2.265	61.0916	16.6345	0.2723
2.270	61.0356	16.6152	0.2722
2.275	60.9910	16.6646	0.2732
2.280	60.9699	16.6763	0.2735
2.285	60.9857	16.7433	0.2745
2.290	60.9328	16.7423	0.2748
2.295	60.9445	16.7380	0.2746
2.300	60.8859	16.7320	0.2748
2.305	60.8511	16.8508	0.2769
2.310	60.8267	16.7788	0.2758
2.315	60.8589	16.8279	0.2765
2.320	60.8052	16.8893	0.2778
2.325	60.7368	16.8612	0.2776
2.330	60.7933	16.9688	0.2791
2.335	60.7111	16.9677	0.2795
2.340	60.7360	16.9355	0.2788
2.345	60.6568	17.0124	0.2805
2.350	60.6574	17.0535	0.2811
2.355	60.6115	17.0129	0.2807
2.360	60.6345	17.0726	0.2816
2.365	60.6061	17.1214	0.2825
2.370	60.5824	17.1052	0.2823
2.375	60.5485	17.0986	0.2824
2.380	60.5696	17.1563	0.2832
2.385	60.5265	17.2013	0.2842

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.390	60.5413	17.2021	0.2841
2.395	60.4954	17.2762	0.2856
2.400	60.4418	17.2936	0.2861
2.405	60.4114	17.2486	0.2855
2.410	60.4416	17.2980	0.2862
2.415	60.3868	17.3353	0.2871
2.420	60.3976	17.3752	0.2877
2.425	60.3463	17.3635	0.2877
2.430	60.3234	17.3612	0.2878
2.435	60.3338	17.3832	0.2881
2.440	60.3088	17.4101	0.2887
2.445	60.2958	17.4398	0.2892
2.450	60.3082	17.4567	0.2895
2.455	60.2431	17.4692	0.2900
2.460	60.2500	17.4984	0.2904
2.465	60.2566	17.5423	0.2911
2.470	60.1856	17.5325	0.2913
2.475	60.1937	17.5639	0.2918
2.480	60.1638	17.5950	0.2925
2.485	60.1782	17.6236	0.2929
2.490	60.1483	17.6282	0.2931
2.495	60.1265	17.6734	0.2939
2.500	60.1268	17.6717	0.2939
2.505	60.0777	17.6923	0.2945
2.510	60.0794	17.7261	0.2950
2.515	60.0631	17.7313	0.2952
2.520	60.0520	17.7735	0.2960
2.525	59.9879	17.8046	0.2968
2.530	59.9843	17.8121	0.2969
2.535	60.0126	17.8235	0.2970
2.540	59.9617	17.8335	0.2974
2.545	59.9372	17.8438	0.2977
2.550	59.9235	17.8834	0.2984
2.555	59.9173	17.8901	0.2986
2.560	59.8786	17.9406	0.2996
2.565	59.8516	17.9543	0.3000
2.570	59.8312	17.9488	0.3000
2.575	59.8333	17.9824	0.3005
2.580	59.7866	17.9977	0.3010
2.585	59.7856	18.0740	0.3023
2.590	59.7443	18.0322	0.3018
2.595	59.7348	18.0920	0.3029
2.600	59.6752	18.1042	0.3034
2.605	59.6262	18.1150	0.3038
2.610	59.6766	18.1257	0.3037
2.615	59.6162	18.1285	0.3041

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.620	59.5929	18.1816	0.3051
2.625	59.5614	18.2020	0.3056
2.630	59.5697	18.2350	0.3061
2.635	59.5519	18.2346	0.3062
2.640	59.4838	18.2913	0.3075
2.645	59.4933	18.2878	0.3074
2.650	59.4513	18.3489	0.3086
2.655	59.4468	18.3272	0.3083
2.660	59.3964	18.3636	0.3092
2.665	59.4184	18.3606	0.3090
2.670	59.3645	18.3512	0.3091
2.675	59.3441	18.3945	0.3100
2.680	59.3236	18.4079	0.3103
2.685	59.2829	18.4406	0.3111
2.690	59.2608	18.4587	0.3115
2.695	59.2623	18.4852	0.3119
2.700	59.2614	18.5279	0.3126
2.705	59.2329	18.5341	0.3129
2.710	59.2330	18.6168	0.3143
2.715	59.2181	18.5844	0.3138
2.720	59.1906	18.6530	0.3151
2.725	59.1264	18.6738	0.3158
2.730	59.1048	18.7291	0.3169
2.735	59.1221	18.7854	0.3177
2.740	59.0951	18.6992	0.3164
2.745	59.0960	18.8246	0.3185
2.750	59.0547	18.8441	0.3191
2.755	59.0372	18.8782	0.3198
2.760	59.0504	18.8872	0.3198
2.765	58.9753	18.9118	0.3207
2.770	59.0164	18.9726	0.3215
2.775	58.9346	18.9698	0.3219
2.780	58.9341	18.9690	0.3219
2.785	58.9375	19.0222	0.3228
2.790	58.9087	19.0074	0.3227
2.795	58.9061	19.0848	0.3240
2.800	58.8851	19.1019	0.3244
2.805	58.8746	19.1237	0.3248
2.810	58.7911	19.1433	0.3256
2.815	58.8258	19.1878	0.3262
2.820	58.8084	19.1802	0.3261
2.825	58.7753	19.2010	0.3267
2.830	58.7696	19.2465	0.3275
2.835	58.7683	19.2618	0.3278
2.840	58.7269	19.3296	0.3291
2.845	58.7184	19.3497	0.3295

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.850	58.6747	19.2983	0.3289
2.855	58.6962	19.3708	0.3300
2.860	58.6459	19.3834	0.3305
2.865	58.6003	19.4394	0.3317
2.870	58.5838	19.4421	0.3319
2.875	58.5630	19.4664	0.3324
2.880	58.5655	19.5209	0.3333
2.885	58.5530	19.5286	0.3335
2.890	58.5161	19.6235	0.3354
2.895	58.4841	19.5871	0.3349
2.900	58.4945	19.5853	0.3348
2.905	58.4899	19.6558	0.3361
2.910	58.4914	19.7030	0.3369
2.915	58.4549	19.6973	0.3370
2.920	58.4507	19.7149	0.3373
2.925	58.4251	19.6803	0.3368
2.930	58.3896	19.7307	0.3379
2.935	58.3683	19.7577	0.3385
2.940	58.3904	19.7648	0.3385
2.945	58.3448	19.8168	0.3396
2.950	58.2947	19.8311	0.3402
2.955	58.3651	19.7813	0.3389
2.960	58.2295	19.9525	0.3427
2.965	58.2576	19.8425	0.3406
2.970	58.1782	19.8643	0.3414
2.975	58.1744	19.9021	0.3421
2.980	58.2034	19.9250	0.3423
2.985	58.1095	19.9611	0.3435
2.990	58.1195	20.0290	0.3446
2.995	58.0717	19.9814	0.3441
3.000	58.0437	20.0807	0.3460
3.005	58.0416	20.0422	0.3453
3.010	57.9880	20.0089	0.3451
3.015	57.9865	20.0726	0.3462
3.020	57.9745	20.1970	0.3484
3.025	57.9554	20.1440	0.3476
3.030	57.9392	20.1457	0.3477
3.035	57.9669	20.2140	0.3487
3.040	57.9146	20.2805	0.3502
3.045	57.8717	20.2799	0.3504
3.050	57.7873	20.2774	0.3509
3.055	57.8694	20.3800	0.3522
3.060	57.8812	20.4308	0.3530
3.065	57.9107	20.3545	0.3515
3.070	57.8961	20.4783	0.3537
3.075	57.9033	20.4732	0.3536

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.080	57.8873	20.4896	0.3540
3.085	57.8385	20.4646	0.3538
3.090	57.8854	20.4997	0.3541
3.095	57.8778	20.4936	0.3541
3.100	57.8727	20.4994	0.3542
3.105	57.8141	20.4899	0.3544
3.110	57.8304	20.5299	0.3550
3.115	57.8241	20.5141	0.3548
3.120	57.8026	20.5401	0.3553
3.125	57.7455	20.5560	0.3560
3.130	57.7338	20.5529	0.3560
3.135	57.6657	20.5983	0.3572
3.140	57.6905	20.5977	0.3570
3.145	57.6142	20.5870	0.3573
3.150	57.6149	20.6330	0.3581
3.155	57.5847	20.6643	0.3589
3.160	57.5950	20.6575	0.3587
3.165	57.5213	20.7403	0.3606
3.170	57.5429	20.6689	0.3592
3.175	57.4863	20.7381	0.3607
3.180	57.4767	20.7906	0.3617
3.185	57.4261	20.8259	0.3627
3.190	57.4290	20.8521	0.3631
3.195	57.4049	20.9170	0.3644
3.200	57.3428	20.9332	0.3651
3.205	57.3935	20.9913	0.3657
3.210	57.3548	20.9921	0.3660
3.215	57.3483	20.9929	0.3661
3.220	57.4221	20.9842	0.3654
3.225	57.3626	21.0815	0.3675
3.230	57.3157	20.9892	0.3662
3.235	57.3301	21.0676	0.3675
3.240	57.2648	21.1171	0.3688
3.245	57.2584	21.0812	0.3682
3.250	57.2299	21.2263	0.3709
3.255	57.1608	21.1996	0.3709
3.260	57.1901	21.1722	0.3702
3.265	57.1753	21.2455	0.3716
3.270	57.0364	21.1666	0.3711
3.275	57.0329	21.2205	0.3721
3.280	57.0768	21.2218	0.3718
3.285	56.9495	21.3655	0.3752
3.290	56.9844	21.2264	0.3725
3.295	56.9041	21.3202	0.3747
3.300	56.9550	21.2860	0.3737
3.305	56.9419	21.4096	0.3760

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.310	56.8553	21.3683	0.3758
3.315	56.7954	21.3828	0.3765
3.320	56.8430	21.4488	0.3773
3.325	56.8182	21.3997	0.3766
3.330	56.7796	21.5028	0.3787
3.335	56.6565	21.5098	0.3797
3.340	56.7004	21.5449	0.3800
3.345	56.5993	21.5541	0.3808
3.350	56.5477	21.6230	0.3824
3.355	56.5711	21.5796	0.3815
3.360	56.5464	21.6863	0.3835
3.365	56.5239	21.7007	0.3839
3.370	56.4371	21.7494	0.3854
3.375	56.5232	21.7434	0.3847
3.380	56.4775	21.7708	0.3855
3.385	56.4162	21.8130	0.3866
3.390	56.3955	21.8270	0.3870
3.395	56.4046	21.8184	0.3868
3.400	56.3319	21.9452	0.3896
3.405	56.3427	21.9076	0.3888
3.410	56.3250	21.9271	0.3893
3.415	56.2700	21.9944	0.3909
3.420	56.2679	21.9789	0.3906
3.425	56.2968	21.9613	0.3901
3.430	56.2921	21.9801	0.3905
3.435	56.2851	22.0729	0.3922
3.440	56.1978	22.0110	0.3917
3.445	56.1759	22.0285	0.3921
3.450	56.0823	22.1207	0.3944
3.455	56.1516	22.1232	0.3940
3.460	56.0990	22.1401	0.3947
3.465	56.0438	22.1471	0.3952
3.470	56.0393	22.1255	0.3948
3.475	56.0251	22.0984	0.3944
3.480	55.9353	22.0855	0.3948
3.485	55.8859	22.0480	0.3945
3.490	55.9101	22.2209	0.3974
3.495	55.8714	22.2313	0.3979
3.500	55.8133	22.1943	0.3977
3.505	55.8817	22.2384	0.3980
3.510	55.7393	22.1934	0.3982
3.515	55.7018	22.3102	0.4005
3.520	55.6349	22.3528	0.4018
3.525	55.5247	22.5263	0.4057
3.530	55.6145	22.5790	0.4060
3.535	55.5406	22.5206	0.4055

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.540	55.5018	22.6769	0.4086
3.545	55.6704	22.7054	0.4079
3.550	55.5919	22.8080	0.4103
3.555	55.6434	22.7430	0.4087
3.560	55.7093	22.8027	0.4093
3.565	55.7546	22.7404	0.4079
3.570	55.7129	22.8739	0.4106
3.575	55.8496	23.0026	0.4119
3.580	55.9421	22.8219	0.4080
3.585	55.9079	22.6047	0.4043
3.590	55.8044	22.4777	0.4028
3.595	55.7465	22.3980	0.4018
3.600	55.7100	22.4972	0.4038
3.605	55.5817	22.3675	0.4024
3.610	55.4792	22.3935	0.4036
3.615	55.3444	22.4728	0.4061
3.620	55.2405	22.5030	0.4074
3.625	55.2918	22.3854	0.4049
3.630	54.9724	22.6468	0.4120
3.635	55.0744	22.7527	0.4131
3.640	55.1335	22.7967	0.4135
3.645	55.1092	22.9220	0.4159
3.650	55.0520	22.9922	0.4176
3.655	55.0327	22.9733	0.4174
3.660	55.0794	23.0925	0.4193
3.665	55.0548	23.0816	0.4192
3.670	55.0394	23.1406	0.4204
3.675	55.0440	23.1149	0.4199
3.680	55.0798	23.1730	0.4207
3.685	55.1518	23.2542	0.4216
3.690	55.1379	23.2576	0.4218
3.695	55.1793	23.2709	0.4217
3.700	55.2086	23.2508	0.4211
3.705	55.2194	23.1911	0.4200
3.710	55.1960	23.2407	0.4211
3.715	55.2456	23.1625	0.4193
3.720	55.2641	23.0623	0.4173
3.725	55.2124	23.1279	0.4189
3.730	55.1611	23.0373	0.4176
3.735	55.0455	23.0127	0.4181
3.740	55.0231	22.9822	0.4177
3.745	54.9484	22.9897	0.4184
3.750	54.8245	23.0558	0.4205
3.755	54.7459	23.1401	0.4227
3.760	54.6755	23.2331	0.4249
3.765	54.6322	23.4094	0.4285

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.770	54.7466	23.4974	0.4292
3.775	54.7585	23.4428	0.4281
3.780	54.8012	23.8862	0.4359
3.785	54.8719	23.6030	0.4301
3.790	54.8675	23.4097	0.4267
3.795	54.8356	23.4374	0.4274
3.800	54.8614	23.5305	0.4289
3.805	54.8523	23.4282	0.4271
3.810	54.7695	23.4817	0.4287
3.815	54.6411	23.4337	0.4289
3.820	54.7307	23.4726	0.4289
3.825	54.6336	23.4462	0.4292
3.830	54.6378	23.4913	0.4299
3.835	53.7102	23.2315	0.4325
3.840	54.5056	23.5266	0.4316
3.845	54.4936	23.5707	0.4325
3.850	54.4843	23.5516	0.4323
3.855	54.4437	23.6193	0.4338
3.860	54.3620	23.5887	0.4339
3.865	54.3340	23.6774	0.4358
3.870	54.3733	23.6470	0.4349
3.875	54.3139	23.6888	0.4361
3.880	54.3106	23.6679	0.4358
3.885	54.2422	23.7391	0.4377
3.890	54.3083	23.7423	0.4372
3.895	54.2291	23.6785	0.4366
3.900	54.1926	23.7341	0.4380
3.905	54.1721	23.8480	0.4402
3.910	54.1778	23.8011	0.4393
3.915	54.1045	23.8245	0.4403
3.920	54.1144	23.8735	0.4412
3.925	54.0521	23.8354	0.4410
3.930	54.0435	23.8567	0.4414
3.935	54.0237	23.8312	0.4411
3.940	54.0091	23.9004	0.4425
3.945	53.9399	23.8792	0.4427
3.950	53.9122	23.9179	0.4436
3.955	53.9084	23.9727	0.4447
3.960	53.8614	23.9537	0.4447
3.965	53.8679	23.9419	0.4445
3.970	53.8437	24.0302	0.4463
3.975	53.8096	24.0468	0.4469
3.980	53.7491	24.0208	0.4469
3.985	53.7409	24.0691	0.4479
3.990	53.7394	24.0609	0.4477
3.995	53.6891	24.0784	0.4485

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.000	53.6702	24.1178	0.4494
4.005	53.6386	24.1171	0.4496
4.010	53.5707	24.1574	0.4509
4.015	53.6162	24.2085	0.4515
4.020	53.5373	24.1784	0.4516
4.025	53.5752	24.1453	0.4507
4.030	53.4809	24.2286	0.4530
4.035	53.4639	24.2567	0.4537
4.040	53.4690	24.2425	0.4534
4.045	53.4323	24.2320	0.4535
4.050	53.3909	24.2276	0.4538
4.055	53.3393	24.2731	0.4551
4.060	53.3126	24.3080	0.4560
4.065	53.2750	24.3536	0.4571
4.070	53.2923	24.3688	0.4573
4.075	53.2300	24.3600	0.4576
4.080	53.2337	24.4254	0.4588
4.085	53.1947	24.4062	0.4588
4.090	53.1894	24.4147	0.4590
4.095	53.1727	24.4782	0.4604
4.100	53.1770	24.4727	0.4602
4.105	53.0260	24.4049	0.4602
4.110	53.1196	24.4817	0.4609
4.115	53.0421	24.5139	0.4622
4.120	53.0125	24.5486	0.4631
4.125	53.0591	24.4957	0.4617
4.130	53.0707	24.5575	0.4627
4.135	52.9712	24.5833	0.4641
4.140	52.9189	24.5221	0.4634
4.145	52.9741	24.5217	0.4629
4.150	52.8489	24.6390	0.4662
4.155	52.8158	24.6878	0.4674
4.160	52.8513	24.6588	0.4666
4.165	52.7549	24.6780	0.4678
4.170	52.7055	24.6749	0.4682
4.175	52.7356	24.6171	0.4668
4.180	52.7375	24.6527	0.4675
4.185	52.6496	24.6171	0.4676
4.190	52.6132	24.6664	0.4688
4.195	52.5967	24.6963	0.4695
4.200	52.6526	24.6891	0.4689
4.205	52.6514	24.7080	0.4693
4.210	52.5345	24.7975	0.4720
4.215	52.6280	24.7831	0.4709
4.220	52.6127	24.8072	0.4715
4.225	52.6677	24.7842	0.4706

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.230	52.5850	24.8309	0.4722
4.235	52.5385	24.8778	0.4735
4.240	52.6093	24.9271	0.4738
4.245	52.5802	24.8947	0.4735
4.250	52.5340	24.9172	0.4743
4.255	52.5691	24.9771	0.4751
4.260	52.5083	24.9899	0.4759
4.265	52.5248	24.9551	0.4751
4.270	52.4906	25.0172	0.4766
4.275	52.4700	25.0591	0.4776
4.280	52.4663	25.0794	0.4780
4.285	52.4580	25.1425	0.4793
4.290	52.4287	25.1321	0.4794
4.295	52.4253	25.1291	0.4793
4.300	52.4217	25.1875	0.4805
4.305	52.3926	25.1997	0.4810
4.310	52.3782	25.1986	0.4811
4.315	52.3748	25.2273	0.4817
4.320	52.3738	25.2310	0.4817
4.325	52.3469	25.2711	0.4828
4.330	52.3381	25.2504	0.4824
4.335	52.3235	25.2825	0.4832
4.340	52.3075	25.2670	0.4830
4.345	52.2765	25.3021	0.4840
4.350	52.2595	25.3178	0.4845
4.355	52.2243	25.3397	0.4852
4.360	52.1887	25.3374	0.4855
4.365	52.1469	25.3891	0.4869
4.370	52.1606	25.3471	0.4859
4.375	52.1057	25.4528	0.4885
4.380	52.0862	25.4262	0.4882
4.385	51.9883	25.4753	0.4900
4.390	51.9922	25.4409	0.4893
4.395	51.9679	25.5104	0.4909
4.400	51.9384	25.5576	0.4921
4.405	51.8663	25.5744	0.4931
4.410	51.9131	25.5963	0.4931
4.415	51.8616	25.5992	0.4936
4.420	51.8566	25.6783	0.4952
4.425	51.8835	25.6281	0.4940
4.430	51.8040	25.6511	0.4952
4.435	51.7214	25.6580	0.4961
4.440	51.7571	25.6584	0.4957
4.445	51.6738	25.7228	0.4978
4.450	51.6516	25.6539	0.4967
4.455	51.6041	25.7607	0.4992

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.460	51.5968	25.6465	0.4971
4.465	51.4969	25.6186	0.4975
4.470	51.5611	25.6081	0.4967
4.475	51.5483	25.5734	0.4961
4.480	51.5699	25.5775	0.4960
4.485	51.4082	25.5802	0.4976
4.490	51.4814	25.6169	0.4976
4.495	51.4412	25.6140	0.4979
4.500	51.4239	25.6527	0.4988
4.505	51.3444	25.6836	0.5002
4.510	51.2864	25.7293	0.5017
4.515	51.3391	25.8031	0.5026
4.520	51.2982	25.8208	0.5033
4.525	51.3414	25.8255	0.5030
4.530	51.1973	25.8269	0.5045
4.535	51.2171	25.8835	0.5054
4.540	51.2022	25.8524	0.5049
4.545	51.1667	25.8543	0.5053
4.550	51.1313	25.8474	0.5055
4.555	51.0992	25.9095	0.5070
4.560	51.0795	25.9036	0.5071
4.565	51.1024	25.9525	0.5079
4.570	51.0611	25.9237	0.5077
4.575	51.0288	25.9002	0.5076
4.580	51.0296	25.9531	0.5086
4.585	50.9691	25.9833	0.5098
4.590	50.9614	25.9780	0.5098
4.595	50.9346	26.0244	0.5109
4.600	50.8989	26.0212	0.5112
4.605	50.8660	26.0544	0.5122
4.610	50.8679	26.0680	0.5125
4.615	50.8086	26.0148	0.5120
4.620	50.7772	26.0031	0.5121
4.625	50.7380	26.0200	0.5128
4.630	50.7280	26.0555	0.5136
4.635	50.6799	26.0428	0.5139
4.640	50.6577	26.0696	0.5146
4.645	50.6350	26.0684	0.5148
4.650	50.5937	26.0880	0.5156
4.655	50.5041	26.0532	0.5159
4.660	50.5906	26.0745	0.5154
4.665	50.5734	26.0469	0.5150
4.670	50.4335	26.1538	0.5186
4.675	50.3688	26.0762	0.5177
4.680	50.3979	26.0856	0.5176
4.685	50.3716	26.1126	0.5184

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.690	50.3911	26.1773	0.5195
4.695	50.3282	26.0706	0.5180
4.700	50.3171	26.1714	0.5201
4.705	50.1440	26.0646	0.5198
4.710	50.1837	26.3000	0.5241
4.715	50.2297	26.2467	0.5225
4.720	50.1848	26.2083	0.5222
4.725	50.2586	26.3172	0.5236
4.730	50.2118	26.2518	0.5228
4.735	50.2479	26.3003	0.5234
4.740	50.2260	26.2939	0.5235
4.745	50.1440	26.3997	0.5265
4.750	50.2146	26.2726	0.5232
4.755	50.2314	26.3658	0.5249
4.760	50.1486	26.3621	0.5257
4.765	50.1923	26.3119	0.5242
4.770	50.1994	26.3834	0.5256
4.775	49.9897	26.4102	0.5283
4.780	50.0297	26.3485	0.5267
4.785	50.1450	26.5479	0.5294
4.790	49.9790	26.4076	0.5284
4.795	49.9131	26.4732	0.5304
4.800	49.9096	26.4772	0.5305
4.805	49.8648	26.4352	0.5301
4.810	49.9179	26.5717	0.5323
4.815	49.8419	26.4983	0.5316
4.820	49.8370	26.6404	0.5345
4.825	49.8448	26.5908	0.5335
4.830	49.7725	26.6918	0.5363
4.835	49.7873	26.6985	0.5363
4.840	49.7944	26.6723	0.5356
4.845	49.7942	26.6449	0.5351
4.850	49.7411	26.7408	0.5376
4.855	49.8278	26.7023	0.5359
4.860	49.7304	26.7365	0.5376
4.865	49.7581	26.6578	0.5357
4.870	49.7118	26.7270	0.5376
4.875	49.7071	26.7509	0.5382
4.880	49.7179	26.7784	0.5386
4.885	49.6223	26.7643	0.5394
4.890	49.7277	26.7637	0.5382
4.895	49.6034	26.7770	0.5398
4.900	49.6376	26.7913	0.5397
4.905	49.5976	26.8066	0.5405
4.910	49.5460	26.8087	0.5411
4.915	49.5446	26.8211	0.5414

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.920	49.5235	26.8080	0.5413
4.925	49.5200	26.8122	0.5414
4.930	49.4867	26.8294	0.5422
4.935	49.4812	26.8689	0.5430
4.940	49.4817	26.8523	0.5427
4.945	49.4574	26.8985	0.5439
4.950	49.4887	26.9088	0.5437
4.955	49.4889	26.9350	0.5443
4.960	49.4323	26.9662	0.5455
4.965	49.4454	26.9884	0.5458
4.970	49.4351	27.0381	0.5469
4.975	49.4282	27.0165	0.5466
4.980	49.3663	27.1143	0.5492
4.985	49.3273	27.1132	0.5497
4.990	49.3160	27.1720	0.5510
4.995	49.2331	27.2458	0.5534
5.000	49.1955	27.2351	0.5536
5.005	49.1457	27.2660	0.5548
5.010	49.0815	27.2684	0.5556
5.015	49.0730	27.3181	0.5567
5.020	49.0052	27.3155	0.5574
5.025	48.9088	27.3737	0.5597
5.030	48.8366	27.3877	0.5608
5.035	48.8437	27.3961	0.5609
5.040	48.7587	27.4186	0.5623
5.045	48.7220	27.3597	0.5615
5.050	48.6161	27.3282	0.5621
5.055	48.6847	27.3092	0.5609
5.060	48.5617	27.3229	0.5626
5.065	48.5215	27.2779	0.5622
5.070	48.5517	27.2864	0.5620
5.075	48.5189	27.2454	0.5615
5.080	48.4480	27.2509	0.5625
5.085	48.4302	27.2219	0.5621
5.090	48.4835	27.2380	0.5618
5.095	48.5169	27.2654	0.5620
5.100	48.5162	27.2388	0.5614
5.105	48.5531	27.3593	0.5635
5.110	48.4486	27.2744	0.5630
5.115	48.4323	27.2891	0.5634
5.120	48.4165	27.3077	0.5640
5.125	48.3997	27.3123	0.5643
5.130	48.3757	27.3256	0.5649
5.135	48.3543	27.3617	0.5659
5.140	48.3343	27.3682	0.5662
5.145	48.2087	27.3146	0.5666

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.150	48.2821	27.4208	0.5679
5.155	48.2247	27.3717	0.5676
5.160	48.1920	27.3721	0.5680
5.165	48.1820	27.4106	0.5689
5.170	48.1113	27.3980	0.5695
5.175	48.0867	27.3740	0.5693
5.180	48.0996	27.3794	0.5692
5.185	48.0615	27.4615	0.5714
5.190	48.0354	27.4020	0.5705
5.195	48.0370	27.3967	0.5703
5.200	48.0412	27.3920	0.5702
5.205	48.0162	27.3960	0.5706
5.210	48.0125	27.3899	0.5705
5.215	48.0032	27.4160	0.5711
5.220	47.9474	27.4259	0.5720
5.225	47.9080	27.3959	0.5718
5.230	47.9252	27.4099	0.5719
5.235	47.8959	27.4654	0.5734
5.240	47.8556	27.4601	0.5738
5.245	47.8353	27.4586	0.5740
5.250	47.8353	27.4943	0.5748
5.255	47.7727	27.5262	0.5762
5.260	47.8212	27.4680	0.5744
5.265	47.8323	27.4946	0.5748
5.270	47.7369	27.5112	0.5763
5.275	47.6814	27.4970	0.5767
5.280	47.7236	27.5327	0.5769
5.285	47.7040	27.4807	0.5761
5.290	47.6330	27.5022	0.5774
5.295	47.6210	27.4344	0.5761
5.300	47.5987	27.5159	0.5781
5.305	47.6243	27.4659	0.5767
5.310	47.4932	27.5683	0.5805
5.315	47.5429	27.5556	0.5796
5.320	47.4553	27.5298	0.5801
5.325	47.4856	27.5447	0.5801
5.330	47.4459	27.5707	0.5811
5.335	47.3886	27.6288	0.5830
5.340	47.3975	27.6751	0.5839
5.345	47.3544	27.7055	0.5851
5.350	47.3911	27.5940	0.5823
5.355	47.3184	27.6814	0.5850
5.360	47.2412	27.6009	0.5843
5.365	47.3555	27.6332	0.5835
5.370	47.1804	27.6725	0.5865
5.375	47.2804	27.6777	0.5854

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.380	47.2677	27.6804	0.5856
5.385	47.1878	27.7028	0.5871
5.390	47.2149	27.7960	0.5887
5.395	47.2586	27.7502	0.5872
5.400	47.1744	27.7783	0.5888
5.405	47.1293	27.7612	0.5890
5.410	47.1167	27.8023	0.5901
5.415	47.0944	27.7561	0.5894
5.420	47.0969	27.8418	0.5912
5.425	47.0927	27.8892	0.5922
5.430	47.0504	27.8252	0.5914
5.435	47.0258	27.8633	0.5925
5.440	46.9859	27.9684	0.5953
5.445	47.0246	27.8990	0.5933
5.450	46.9595	27.9332	0.5948
5.455	47.0117	27.9194	0.5939
5.460	46.9373	27.9407	0.5953
5.465	47.0096	28.0042	0.5957
5.470	46.8659	28.0535	0.5986
5.475	46.8991	28.0184	0.5974
5.480	46.8780	28.0836	0.5991
5.485	46.8404	28.0397	0.5986
5.490	46.8474	28.1562	0.6010
5.495	46.8690	28.0612	0.5987
5.500	46.8629	28.1110	0.5999
5.505	46.8754	28.1077	0.5996
5.510	46.8196	28.1187	0.6006
5.515	46.8141	28.0898	0.6000
5.520	46.7508	28.1352	0.6018
5.525	46.7800	28.1077	0.6008
5.530	46.7571	28.1159	0.6013
5.535	46.7420	28.0952	0.6011
5.540	46.7255	28.1498	0.6025
5.545	46.6893	28.1542	0.6030
5.550	46.6783	28.1535	0.6031
5.555	46.6567	28.1406	0.6031
5.560	46.6420	28.1743	0.6041
5.565	46.6647	28.1590	0.6034
5.570	46.6419	28.1655	0.6039
5.575	46.6491	28.1649	0.6038
5.580	46.6512	28.2020	0.6045
5.585	46.6321	28.1934	0.6046
5.590	46.6181	28.1701	0.6043
5.595	46.5943	28.1716	0.6046
5.600	46.6013	28.1779	0.6047
5.605	46.5866	28.1430	0.6041

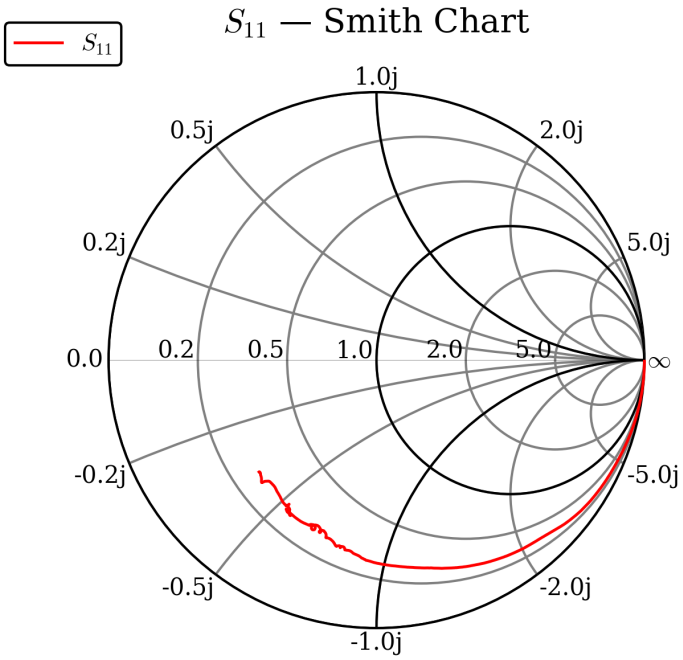
(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.610	46.5048	28.1312	0.6049
5.615	46.4687	28.1247	0.6052
5.620	46.4510	28.1278	0.6055
5.625	46.4647	28.1702	0.6063
5.630	46.3819	28.1503	0.6069
5.635	46.3682	28.1260	0.6066
5.640	46.3010	28.1227	0.6074
5.645	46.2615	28.1420	0.6083
5.650	46.3015	28.1658	0.6083
5.655	46.2378	28.1662	0.6092
5.660	46.2158	28.1797	0.6097
5.665	46.2706	28.1303	0.6080
5.670	46.1211	28.2847	0.6133
5.675	46.1175	28.1771	0.6110
5.680	46.0634	28.1876	0.6119
5.685	46.1035	28.1121	0.6098
5.690	46.0485	28.1801	0.6120
5.695	46.0607	28.2172	0.6126
5.700	45.9647	28.1910	0.6133
5.705	45.8931	28.2545	0.6157
5.710	45.9785	28.1454	0.6121
5.715	45.8462	28.1827	0.6147
5.720	45.8319	28.2220	0.6158
5.725	45.8144	28.2952	0.6176
5.730	45.7851	28.2261	0.6165
5.735	45.7769	28.2393	0.6169
5.740	45.7760	28.2645	0.6175
5.745	45.7093	28.3571	0.6204
5.750	45.6598	28.3257	0.6204
5.755	45.6869	28.2975	0.6194
5.760	45.5972	28.3400	0.6215
5.765	45.5947	28.2753	0.6201
5.770	45.5596	28.2962	0.6211
5.775	45.5868	28.2891	0.6206
5.780	45.5544	28.3837	0.6231
5.785	45.5107	28.3667	0.6233
5.790	45.4765	28.4305	0.6252
5.795	45.4761	28.4151	0.6248
5.800	45.4395	28.4013	0.6250
5.805	45.3894	28.4108	0.6259
5.810	45.4264	28.4458	0.6262
5.815	45.3969	28.4346	0.6264
5.820	45.3511	28.4130	0.6265
5.825	45.2998	28.4672	0.6284
5.830	45.2594	28.4663	0.6290
5.835	45.2728	28.4493	0.6284

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.840	45.2669	28.4630	0.6288
5.845	45.2267	28.4755	0.6296
5.850	45.1910	28.4560	0.6297
5.855	45.1822	28.4776	0.6303
5.860	45.1480	28.5000	0.6313
5.865	45.1603	28.5044	0.6312
5.870	45.0977	28.4812	0.6315
5.875	45.1024	28.5074	0.6321
5.880	45.0666	28.5155	0.6327
5.885	45.0291	28.4593	0.6320
5.890	45.0111	28.5960	0.6353
5.895	45.0030	28.5478	0.6344
5.900	44.9663	28.5110	0.6341
5.905	45.0062	28.5903	0.6353
5.910	44.8704	28.5339	0.6359
5.915	44.9079	28.5562	0.6359
5.920	44.8685	28.5323	0.6359
5.925	44.8166	28.5277	0.6365
5.930	44.7741	28.6057	0.6389
5.935	44.7238	28.5684	0.6388
5.940	44.7275	28.5951	0.6393
5.945	44.7896	28.5350	0.6371
5.950	44.7216	28.6380	0.6404
5.955	44.7321	28.6345	0.6401
5.960	44.6846	28.6437	0.6410
5.965	44.6538	28.6628	0.6419
5.970	44.6576	28.6711	0.6420
5.975	44.5678	28.5151	0.6398
5.980	44.7420	28.5713	0.6386
5.985	44.5611	28.7240	0.6446
5.990	44.4842	28.6324	0.6437
5.995	44.4964	28.6424	0.6437
6.000	44.5269	28.5634	0.6415

1.2 S_{11} — Smith Chart



Frequency (GHz)	Re(S_{11})	Im(S_{11})	$\lvert S_{11}\rvert$ (dB)
0.005	0.9998	-0.0042	-0.00
0.010	0.9999	-0.0084	-0.00
0.015	0.9998	-0.0126	-0.00
0.020	0.9996	-0.0170	-0.00
0.025	0.9997	-0.0213	-0.00
0.030	0.9995	-0.0256	-0.00
0.035	0.9993	-0.0298	-0.00
0.040	0.9991	-0.0341	-0.00
0.045	0.9989	-0.0383	-0.00
0.050	0.9987	-0.0425	-0.00
0.055	0.9985	-0.0470	-0.00
0.060	0.9981	-0.0513	-0.00
0.065	0.9979	-0.0555	-0.00
0.070	0.9975	-0.0596	-0.01
0.075	0.9972	-0.0639	-0.01
0.080	0.9968	-0.0681	-0.01
0.085	0.9964	-0.0722	-0.01
0.090	0.9959	-0.0766	-0.01
0.095	0.9954	-0.0808	-0.01
0.100	0.9951	-0.0849	-0.01
0.105	0.9945	-0.0892	-0.01
0.110	0.9939	-0.0933	-0.02
0.115	0.9935	-0.0977	-0.01

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.120	0.9929	-0.1018	-0.02
0.125	0.9924	-0.1060	-0.02
0.130	0.9917	-0.1102	-0.02
0.135	0.9910	-0.1144	-0.02
0.140	0.9903	-0.1186	-0.02
0.145	0.9898	-0.1227	-0.02
0.150	0.9890	-0.1268	-0.02
0.155	0.9884	-0.1310	-0.03
0.160	0.9876	-0.1351	-0.03
0.165	0.9868	-0.1393	-0.03
0.170	0.9861	-0.1433	-0.03
0.175	0.9853	-0.1475	-0.03
0.180	0.9844	-0.1516	-0.03
0.185	0.9834	-0.1558	-0.04
0.190	0.9826	-0.1599	-0.04
0.195	0.9817	-0.1640	-0.04
0.200	0.9808	-0.1681	-0.04
0.205	0.9799	-0.1723	-0.04
0.210	0.9789	-0.1763	-0.05
0.215	0.9779	-0.1804	-0.05
0.220	0.9770	-0.1845	-0.05
0.225	0.9757	-0.1888	-0.05
0.230	0.9748	-0.1927	-0.06
0.235	0.9735	-0.1968	-0.06
0.240	0.9725	-0.2008	-0.06
0.245	0.9713	-0.2050	-0.06
0.250	0.9701	-0.2090	-0.07
0.255	0.9691	-0.2130	-0.07
0.260	0.9679	-0.2170	-0.07
0.265	0.9666	-0.2211	-0.07
0.270	0.9653	-0.2251	-0.08
0.275	0.9641	-0.2290	-0.08
0.280	0.9627	-0.2330	-0.08
0.285	0.9614	-0.2370	-0.09
0.290	0.9600	-0.2409	-0.09
0.295	0.9588	-0.2449	-0.09
0.300	0.9574	-0.2488	-0.09
0.305	0.9559	-0.2527	-0.10
0.310	0.9545	-0.2566	-0.10
0.315	0.9532	-0.2605	-0.10
0.320	0.9516	-0.2644	-0.11
0.325	0.9501	-0.2683	-0.11
0.330	0.9487	-0.2721	-0.11
0.335	0.9473	-0.2758	-0.12
0.340	0.9456	-0.2798	-0.12
0.345	0.9441	-0.2836	-0.12

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.350	0.9425	-0.2874	-0.13
0.355	0.9409	-0.2911	-0.13
0.360	0.9394	-0.2949	-0.14
0.365	0.9378	-0.2987	-0.14
0.370	0.9361	-0.3024	-0.14
0.375	0.9343	-0.3062	-0.15
0.380	0.9326	-0.3098	-0.15
0.385	0.9310	-0.3135	-0.15
0.390	0.9294	-0.3172	-0.16
0.395	0.9278	-0.3209	-0.16
0.400	0.9260	-0.3244	-0.16
0.405	0.9244	-0.3280	-0.17
0.410	0.9226	-0.3316	-0.17
0.415	0.9211	-0.3353	-0.17
0.420	0.9192	-0.3389	-0.18
0.425	0.9174	-0.3423	-0.18
0.430	0.9157	-0.3459	-0.19
0.435	0.9139	-0.3494	-0.19
0.440	0.9121	-0.3528	-0.19
0.445	0.9104	-0.3563	-0.20
0.450	0.9086	-0.3598	-0.20
0.455	0.9068	-0.3633	-0.20
0.460	0.9050	-0.3667	-0.21
0.465	0.9030	-0.3701	-0.21
0.470	0.9012	-0.3735	-0.22
0.475	0.8993	-0.3769	-0.22
0.480	0.8974	-0.3802	-0.22
0.485	0.8957	-0.3836	-0.23
0.490	0.8938	-0.3868	-0.23
0.495	0.8919	-0.3902	-0.23
0.500	0.8901	-0.3935	-0.24
0.505	0.8883	-0.3968	-0.24
0.510	0.8863	-0.4000	-0.24
0.515	0.8843	-0.4032	-0.25
0.520	0.8826	-0.4064	-0.25
0.525	0.8809	-0.4096	-0.25
0.530	0.8789	-0.4126	-0.26
0.535	0.8771	-0.4159	-0.26
0.540	0.8752	-0.4192	-0.26
0.545	0.8732	-0.4223	-0.27
0.550	0.8712	-0.4255	-0.27
0.555	0.8693	-0.4286	-0.27
0.560	0.8674	-0.4317	-0.27
0.565	0.8654	-0.4348	-0.28
0.570	0.8634	-0.4378	-0.28
0.575	0.8614	-0.4410	-0.28

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.580	0.8594	-0.4441	-0.29
0.585	0.8575	-0.4471	-0.29
0.590	0.8555	-0.4502	-0.29
0.595	0.8535	-0.4533	-0.30
0.600	0.8514	-0.4562	-0.30
0.605	0.8494	-0.4592	-0.30
0.610	0.8473	-0.4622	-0.31
0.615	0.8453	-0.4651	-0.31
0.620	0.8432	-0.4680	-0.32
0.625	0.8412	-0.4710	-0.32
0.630	0.8389	-0.4739	-0.32
0.635	0.8369	-0.4767	-0.33
0.640	0.8347	-0.4797	-0.33
0.645	0.8326	-0.4824	-0.33
0.650	0.8305	-0.4852	-0.34
0.655	0.8284	-0.4880	-0.34
0.660	0.8263	-0.4908	-0.34
0.665	0.8242	-0.4936	-0.35
0.670	0.8221	-0.4963	-0.35
0.675	0.8199	-0.4991	-0.36
0.680	0.8178	-0.5017	-0.36
0.685	0.8156	-0.5044	-0.36
0.690	0.8134	-0.5069	-0.37
0.695	0.8112	-0.5098	-0.37
0.700	0.8089	-0.5124	-0.38
0.705	0.8067	-0.5152	-0.38
0.710	0.8044	-0.5179	-0.38
0.715	0.8022	-0.5206	-0.39
0.720	0.7999	-0.5234	-0.39
0.725	0.7975	-0.5259	-0.40
0.730	0.7952	-0.5285	-0.40
0.735	0.7930	-0.5311	-0.41
0.740	0.7906	-0.5337	-0.41
0.745	0.7881	-0.5363	-0.42
0.750	0.7858	-0.5388	-0.42
0.755	0.7834	-0.5414	-0.42
0.760	0.7810	-0.5439	-0.43
0.765	0.7786	-0.5464	-0.43
0.770	0.7760	-0.5489	-0.44
0.775	0.7735	-0.5515	-0.45
0.780	0.7710	-0.5543	-0.45
0.785	0.7685	-0.5568	-0.45
0.790	0.7658	-0.5593	-0.46
0.795	0.7633	-0.5618	-0.47
0.800	0.7606	-0.5642	-0.47
0.805	0.7580	-0.5668	-0.48

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.810	0.7552	-0.5692	-0.48
0.815	0.7524	-0.5716	-0.49
0.820	0.7498	-0.5740	-0.50
0.825	0.7471	-0.5763	-0.51
0.830	0.7442	-0.5787	-0.51
0.835	0.7415	-0.5808	-0.52
0.840	0.7386	-0.5831	-0.53
0.845	0.7361	-0.5853	-0.53
0.850	0.7333	-0.5875	-0.54
0.855	0.7306	-0.5898	-0.55
0.860	0.7278	-0.5920	-0.55
0.865	0.7251	-0.5943	-0.56
0.870	0.7223	-0.5965	-0.57
0.875	0.7194	-0.5986	-0.58
0.880	0.7165	-0.6009	-0.58
0.885	0.7136	-0.6031	-0.59
0.890	0.7108	-0.6051	-0.60
0.895	0.7077	-0.6074	-0.61
0.900	0.7048	-0.6095	-0.61
0.905	0.7018	-0.6116	-0.62
0.910	0.6988	-0.6135	-0.63
0.915	0.6957	-0.6155	-0.64
0.920	0.6928	-0.6173	-0.65
0.925	0.6897	-0.6192	-0.66
0.930	0.6867	-0.6211	-0.67
0.935	0.6837	-0.6229	-0.68
0.940	0.6808	-0.6247	-0.69
0.945	0.6778	-0.6265	-0.70
0.950	0.6747	-0.6283	-0.71
0.955	0.6718	-0.6300	-0.71
0.960	0.6691	-0.6318	-0.72
0.965	0.6660	-0.6333	-0.73
0.970	0.6633	-0.6352	-0.74
0.975	0.6604	-0.6368	-0.75
0.980	0.6577	-0.6385	-0.76
0.985	0.6550	-0.6401	-0.76
0.990	0.6522	-0.6417	-0.77
0.995	0.6494	-0.6433	-0.78
1.000	0.6467	-0.6449	-0.79
1.005	0.6440	-0.6465	-0.80
1.010	0.6411	-0.6480	-0.80
1.015	0.6384	-0.6496	-0.81
1.020	0.6357	-0.6511	-0.82
1.025	0.6331	-0.6528	-0.83
1.030	0.6302	-0.6543	-0.83
1.035	0.6277	-0.6559	-0.84

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.040	0.6251	-0.6576	-0.85
1.045	0.6225	-0.6591	-0.85
1.050	0.6198	-0.6605	-0.86
1.055	0.6173	-0.6621	-0.86
1.060	0.6147	-0.6636	-0.87
1.065	0.6121	-0.6651	-0.88
1.070	0.6096	-0.6667	-0.88
1.075	0.6071	-0.6683	-0.89
1.080	0.6046	-0.6697	-0.89
1.085	0.6022	-0.6711	-0.90
1.090	0.5997	-0.6728	-0.90
1.095	0.5972	-0.6744	-0.91
1.100	0.5946	-0.6759	-0.91
1.105	0.5921	-0.6772	-0.92
1.110	0.5896	-0.6788	-0.92
1.115	0.5871	-0.6803	-0.93
1.120	0.5847	-0.6818	-0.93
1.125	0.5822	-0.6832	-0.94
1.130	0.5798	-0.6847	-0.94
1.135	0.5772	-0.6861	-0.95
1.140	0.5748	-0.6876	-0.95
1.145	0.5724	-0.6890	-0.96
1.150	0.5700	-0.6903	-0.96
1.155	0.5674	-0.6919	-0.97
1.160	0.5651	-0.6932	-0.97
1.165	0.5627	-0.6945	-0.98
1.170	0.5603	-0.6959	-0.98
1.175	0.5579	-0.6972	-0.98
1.180	0.5557	-0.6984	-0.99
1.185	0.5534	-0.6998	-0.99
1.190	0.5510	-0.7009	-1.00
1.195	0.5489	-0.7022	-1.00
1.200	0.5466	-0.7036	-1.00
1.205	0.5442	-0.7045	-1.01
1.210	0.5419	-0.7057	-1.01
1.215	0.5395	-0.7070	-1.02
1.220	0.5372	-0.7080	-1.02
1.225	0.5349	-0.7091	-1.03
1.230	0.5325	-0.7102	-1.03
1.235	0.5303	-0.7114	-1.04
1.240	0.5279	-0.7125	-1.04
1.245	0.5257	-0.7136	-1.05
1.250	0.5235	-0.7145	-1.05
1.255	0.5212	-0.7156	-1.06
1.260	0.5190	-0.7167	-1.06
1.265	0.5169	-0.7176	-1.07

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.270	0.5146	-0.7185	-1.07
1.275	0.5123	-0.7195	-1.08
1.280	0.5101	-0.7205	-1.08
1.285	0.5077	-0.7216	-1.09
1.290	0.5055	-0.7225	-1.09
1.295	0.5032	-0.7236	-1.10
1.300	0.5010	-0.7245	-1.10
1.305	0.4989	-0.7254	-1.11
1.310	0.4967	-0.7263	-1.11
1.315	0.4946	-0.7272	-1.12
1.320	0.4924	-0.7280	-1.12
1.325	0.4903	-0.7290	-1.13
1.330	0.4880	-0.7298	-1.13
1.335	0.4860	-0.7307	-1.13
1.340	0.4839	-0.7316	-1.14
1.345	0.4814	-0.7322	-1.15
1.350	0.4793	-0.7331	-1.15
1.355	0.4770	-0.7340	-1.16
1.360	0.4748	-0.7349	-1.16
1.365	0.4726	-0.7356	-1.17
1.370	0.4704	-0.7366	-1.17
1.375	0.4681	-0.7373	-1.18
1.380	0.4656	-0.7381	-1.18
1.385	0.4633	-0.7389	-1.19
1.390	0.4609	-0.7397	-1.19
1.395	0.4585	-0.7405	-1.20
1.400	0.4561	-0.7413	-1.21
1.405	0.4535	-0.7421	-1.21
1.410	0.4510	-0.7428	-1.22
1.415	0.4486	-0.7436	-1.22
1.420	0.4460	-0.7445	-1.23
1.425	0.4435	-0.7454	-1.24
1.430	0.4408	-0.7462	-1.24
1.435	0.4384	-0.7470	-1.25
1.440	0.4356	-0.7479	-1.25
1.445	0.4331	-0.7487	-1.26
1.450	0.4303	-0.7494	-1.27
1.455	0.4278	-0.7503	-1.27
1.460	0.4251	-0.7512	-1.28
1.465	0.4223	-0.7518	-1.29
1.470	0.4197	-0.7525	-1.29
1.475	0.4170	-0.7534	-1.30
1.480	0.4142	-0.7540	-1.31
1.485	0.4113	-0.7547	-1.31
1.490	0.4085	-0.7556	-1.32
1.495	0.4056	-0.7564	-1.33

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.500	0.4030	-0.7572	-1.33
1.505	0.4000	-0.7577	-1.34
1.510	0.3971	-0.7583	-1.35
1.515	0.3942	-0.7589	-1.36
1.520	0.3912	-0.7596	-1.37
1.525	0.3883	-0.7604	-1.37
1.530	0.3854	-0.7611	-1.38
1.535	0.3823	-0.7616	-1.39
1.540	0.3793	-0.7622	-1.40
1.545	0.3763	-0.7629	-1.40
1.550	0.3734	-0.7635	-1.41
1.555	0.3701	-0.7639	-1.42
1.560	0.3670	-0.7645	-1.43
1.565	0.3639	-0.7652	-1.44
1.570	0.3607	-0.7656	-1.45
1.575	0.3577	-0.7664	-1.45
1.580	0.3545	-0.7668	-1.47
1.585	0.3515	-0.7675	-1.47
1.590	0.3484	-0.7680	-1.48
1.595	0.3452	-0.7687	-1.49
1.600	0.3422	-0.7692	-1.50
1.605	0.3391	-0.7698	-1.50
1.610	0.3361	-0.7702	-1.51
1.615	0.3330	-0.7706	-1.52
1.620	0.3298	-0.7709	-1.53
1.625	0.3267	-0.7713	-1.54
1.630	0.3235	-0.7718	-1.55
1.635	0.3203	-0.7722	-1.56
1.640	0.3171	-0.7725	-1.57
1.645	0.3141	-0.7728	-1.58
1.650	0.3108	-0.7730	-1.59
1.655	0.3079	-0.7734	-1.59
1.660	0.3049	-0.7738	-1.60
1.665	0.3016	-0.7740	-1.61
1.670	0.2987	-0.7745	-1.62
1.675	0.2956	-0.7745	-1.63
1.680	0.2928	-0.7748	-1.64
1.685	0.2897	-0.7751	-1.65
1.690	0.2870	-0.7751	-1.66
1.695	0.2841	-0.7754	-1.66
1.700	0.2812	-0.7755	-1.67
1.705	0.2784	-0.7756	-1.68
1.710	0.2753	-0.7758	-1.69
1.715	0.2725	-0.7758	-1.70
1.720	0.2696	-0.7760	-1.71
1.725	0.2668	-0.7759	-1.72

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.730	0.2639	-0.7760	-1.73
1.735	0.2612	-0.7762	-1.73
1.740	0.2585	-0.7761	-1.74
1.745	0.2558	-0.7762	-1.75
1.750	0.2533	-0.7764	-1.76
1.755	0.2509	-0.7766	-1.77
1.760	0.2484	-0.7767	-1.77
1.765	0.2459	-0.7769	-1.78
1.770	0.2438	-0.7770	-1.78
1.775	0.2415	-0.7769	-1.79
1.780	0.2394	-0.7767	-1.80
1.785	0.2370	-0.7767	-1.81
1.790	0.2345	-0.7766	-1.82
1.795	0.2323	-0.7764	-1.83
1.800	0.2300	-0.7764	-1.83
1.805	0.2276	-0.7764	-1.84
1.810	0.2256	-0.7765	-1.85
1.815	0.2235	-0.7766	-1.85
1.820	0.2215	-0.7765	-1.86
1.825	0.2197	-0.7764	-1.86
1.830	0.2177	-0.7763	-1.87
1.835	0.2157	-0.7762	-1.88
1.840	0.2139	-0.7762	-1.88
1.845	0.2120	-0.7760	-1.89
1.850	0.2101	-0.7757	-1.90
1.855	0.2082	-0.7755	-1.91
1.860	0.2064	-0.7754	-1.91
1.865	0.2044	-0.7753	-1.92
1.870	0.2026	-0.7752	-1.92
1.875	0.2008	-0.7753	-1.93
1.880	0.1990	-0.7756	-1.93
1.885	0.1975	-0.7756	-1.93
1.890	0.1962	-0.7757	-1.94
1.895	0.1946	-0.7755	-1.94
1.900	0.1930	-0.7753	-1.95
1.905	0.1915	-0.7753	-1.95
1.910	0.1898	-0.7752	-1.96
1.915	0.1882	-0.7753	-1.96
1.920	0.1867	-0.7755	-1.96
1.925	0.1852	-0.7756	-1.97
1.930	0.1837	-0.7755	-1.97
1.935	0.1824	-0.7755	-1.97
1.940	0.1809	-0.7757	-1.98
1.945	0.1796	-0.7756	-1.98
1.950	0.1782	-0.7755	-1.98
1.955	0.1767	-0.7756	-1.99

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.960	0.1754	-0.7754	-1.99
1.965	0.1740	-0.7753	-2.00
1.970	0.1726	-0.7754	-2.00
1.975	0.1714	-0.7753	-2.00
1.980	0.1700	-0.7754	-2.01
1.985	0.1686	-0.7752	-2.01
1.990	0.1672	-0.7751	-2.01
1.995	0.1658	-0.7751	-2.02
2.000	0.1643	-0.7750	-2.02
2.005	0.1630	-0.7750	-2.03
2.010	0.1615	-0.7750	-2.03
2.015	0.1601	-0.7749	-2.03
2.020	0.1587	-0.7749	-2.04
2.025	0.1572	-0.7750	-2.04
2.030	0.1557	-0.7749	-2.04
2.035	0.1541	-0.7748	-2.05
2.040	0.1527	-0.7747	-2.05
2.045	0.1510	-0.7746	-2.06
2.050	0.1496	-0.7746	-2.06
2.055	0.1479	-0.7745	-2.06
2.060	0.1464	-0.7744	-2.07
2.065	0.1446	-0.7743	-2.07
2.070	0.1430	-0.7742	-2.08
2.075	0.1414	-0.7740	-2.08
2.080	0.1395	-0.7740	-2.09
2.085	0.1377	-0.7736	-2.09
2.090	0.1359	-0.7735	-2.10
2.095	0.1339	-0.7733	-2.10
2.100	0.1321	-0.7734	-2.11
2.105	0.1300	-0.7731	-2.11
2.110	0.1280	-0.7732	-2.12
2.115	0.1260	-0.7730	-2.12
2.120	0.1240	-0.7729	-2.13
2.125	0.1219	-0.7730	-2.13
2.130	0.1197	-0.7727	-2.14
2.135	0.1176	-0.7726	-2.14
2.140	0.1154	-0.7727	-2.14
2.145	0.1132	-0.7723	-2.15
2.150	0.1111	-0.7721	-2.16
2.155	0.1088	-0.7719	-2.16
2.160	0.1066	-0.7719	-2.17
2.165	0.1044	-0.7716	-2.17
2.170	0.1021	-0.7714	-2.18
2.175	0.0998	-0.7711	-2.19
2.180	0.0975	-0.7710	-2.19
2.185	0.0950	-0.7709	-2.19

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.190	0.0928	-0.7705	-2.20
2.195	0.0904	-0.7704	-2.21
2.200	0.0879	-0.7702	-2.21
2.205	0.0856	-0.7699	-2.22
2.210	0.0832	-0.7696	-2.22
2.215	0.0809	-0.7693	-2.23
2.220	0.0784	-0.7692	-2.23
2.225	0.0760	-0.7689	-2.24
2.230	0.0735	-0.7685	-2.25
2.235	0.0712	-0.7682	-2.25
2.240	0.0687	-0.7681	-2.26
2.245	0.0662	-0.7679	-2.26
2.250	0.0638	-0.7677	-2.27
2.255	0.0613	-0.7674	-2.27
2.260	0.0590	-0.7672	-2.28
2.265	0.0565	-0.7670	-2.28
2.270	0.0542	-0.7666	-2.29
2.275	0.0517	-0.7663	-2.29
2.280	0.0494	-0.7659	-2.30
2.285	0.0470	-0.7657	-2.30
2.290	0.0447	-0.7653	-2.31
2.295	0.0423	-0.7649	-2.31
2.300	0.0399	-0.7647	-2.32
2.305	0.0376	-0.7642	-2.32
2.310	0.0351	-0.7640	-2.33
2.315	0.0329	-0.7637	-2.33
2.320	0.0306	-0.7631	-2.34
2.325	0.0283	-0.7630	-2.34
2.330	0.0261	-0.7626	-2.35
2.335	0.0240	-0.7621	-2.36
2.340	0.0216	-0.7617	-2.36
2.345	0.0195	-0.7613	-2.37
2.350	0.0173	-0.7609	-2.37
2.355	0.0153	-0.7604	-2.38
2.360	0.0133	-0.7601	-2.38
2.365	0.0113	-0.7595	-2.39
2.370	0.0094	-0.7591	-2.39
2.375	0.0074	-0.7586	-2.40
2.380	0.0054	-0.7585	-2.40
2.385	0.0035	-0.7580	-2.41
2.390	0.0017	-0.7577	-2.41
2.395	-0.0001	-0.7573	-2.41
2.400	-0.0017	-0.7569	-2.42
2.405	-0.0034	-0.7566	-2.42
2.410	-0.0053	-0.7561	-2.43
2.415	-0.0068	-0.7557	-2.43

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.420	-0.0084	-0.7554	-2.44
2.425	-0.0100	-0.7549	-2.44
2.430	-0.0114	-0.7545	-2.45
2.435	-0.0128	-0.7540	-2.45
2.440	-0.0143	-0.7538	-2.45
2.445	-0.0157	-0.7533	-2.46
2.450	-0.0171	-0.7529	-2.46
2.455	-0.0181	-0.7524	-2.47
2.460	-0.0194	-0.7520	-2.47
2.465	-0.0206	-0.7516	-2.48
2.470	-0.0217	-0.7512	-2.48
2.475	-0.0230	-0.7508	-2.49
2.480	-0.0242	-0.7504	-2.49
2.485	-0.0253	-0.7501	-2.49
2.490	-0.0265	-0.7497	-2.50
2.495	-0.0277	-0.7495	-2.50
2.500	-0.0288	-0.7490	-2.50
2.505	-0.0298	-0.7488	-2.51
2.510	-0.0309	-0.7485	-2.51
2.515	-0.0320	-0.7484	-2.51
2.520	-0.0331	-0.7481	-2.51
2.525	-0.0342	-0.7479	-2.51
2.530	-0.0352	-0.7477	-2.52
2.535	-0.0363	-0.7475	-2.52
2.540	-0.0373	-0.7474	-2.52
2.545	-0.0382	-0.7471	-2.52
2.550	-0.0392	-0.7469	-2.52
2.555	-0.0400	-0.7465	-2.53
2.560	-0.0408	-0.7460	-2.53
2.565	-0.0412	-0.7455	-2.54
2.570	-0.0417	-0.7451	-2.54
2.575	-0.0422	-0.7446	-2.55
2.580	-0.0425	-0.7441	-2.55
2.585	-0.0429	-0.7439	-2.56
2.590	-0.0433	-0.7437	-2.56
2.595	-0.0437	-0.7437	-2.56
2.600	-0.0439	-0.7435	-2.56
2.605	-0.0445	-0.7435	-2.56
2.610	-0.0449	-0.7433	-2.56
2.615	-0.0454	-0.7431	-2.56
2.620	-0.0460	-0.7431	-2.56
2.625	-0.0465	-0.7429	-2.56
2.630	-0.0468	-0.7426	-2.57
2.635	-0.0474	-0.7422	-2.57
2.640	-0.0479	-0.7419	-2.58
2.645	-0.0484	-0.7416	-2.58

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.650	-0.0490	-0.7413	-2.58
2.655	-0.0496	-0.7410	-2.58
2.660	-0.0500	-0.7407	-2.59
2.665	-0.0505	-0.7403	-2.59
2.670	-0.0512	-0.7400	-2.59
2.675	-0.0518	-0.7398	-2.60
2.680	-0.0522	-0.7397	-2.60
2.685	-0.0527	-0.7396	-2.60
2.690	-0.0532	-0.7395	-2.60
2.695	-0.0537	-0.7394	-2.60
2.700	-0.0543	-0.7392	-2.60
2.705	-0.0545	-0.7390	-2.60
2.710	-0.0549	-0.7387	-2.61
2.715	-0.0555	-0.7383	-2.61
2.720	-0.0560	-0.7380	-2.61
2.725	-0.0566	-0.7375	-2.62
2.730	-0.0572	-0.7368	-2.63
2.735	-0.0580	-0.7362	-2.63
2.740	-0.0587	-0.7358	-2.64
2.745	-0.0595	-0.7350	-2.65
2.750	-0.0601	-0.7344	-2.65
2.755	-0.0612	-0.7337	-2.66
2.760	-0.0621	-0.7329	-2.67
2.765	-0.0631	-0.7321	-2.68
2.770	-0.0641	-0.7314	-2.68
2.775	-0.0652	-0.7307	-2.69
2.780	-0.0663	-0.7302	-2.70
2.785	-0.0675	-0.7296	-2.70
2.790	-0.0686	-0.7289	-2.71
2.795	-0.0699	-0.7284	-2.71
2.800	-0.0711	-0.7278	-2.72
2.805	-0.0723	-0.7272	-2.72
2.810	-0.0733	-0.7268	-2.73
2.815	-0.0748	-0.7262	-2.73
2.820	-0.0759	-0.7258	-2.74
2.825	-0.0772	-0.7252	-2.74
2.830	-0.0783	-0.7247	-2.75
2.835	-0.0796	-0.7242	-2.75
2.840	-0.0808	-0.7237	-2.75
2.845	-0.0821	-0.7231	-2.76
2.850	-0.0831	-0.7227	-2.76
2.855	-0.0843	-0.7222	-2.77
2.860	-0.0855	-0.7217	-2.77
2.865	-0.0863	-0.7211	-2.78
2.870	-0.0872	-0.7206	-2.78
2.875	-0.0880	-0.7197	-2.79

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.880	-0.0889	-0.7188	-2.80
2.885	-0.0898	-0.7178	-2.81
2.890	-0.0910	-0.7168	-2.82
2.895	-0.0923	-0.7157	-2.83
2.900	-0.0937	-0.7149	-2.84
2.905	-0.0951	-0.7139	-2.85
2.910	-0.0965	-0.7129	-2.86
2.915	-0.0982	-0.7122	-2.87
2.920	-0.0997	-0.7116	-2.87
2.925	-0.1012	-0.7111	-2.87
2.930	-0.1027	-0.7103	-2.88
2.935	-0.1041	-0.7101	-2.88
2.940	-0.1054	-0.7099	-2.88
2.945	-0.1067	-0.7097	-2.88
2.950	-0.1078	-0.7095	-2.88
2.955	-0.1091	-0.7094	-2.88
2.960	-0.1099	-0.7093	-2.88
2.965	-0.1108	-0.7093	-2.88
2.970	-0.1115	-0.7093	-2.88
2.975	-0.1123	-0.7092	-2.88
2.980	-0.1129	-0.7092	-2.88
2.985	-0.1132	-0.7092	-2.88
2.990	-0.1137	-0.7090	-2.88
2.995	-0.1139	-0.7087	-2.88
3.000	-0.1145	-0.7084	-2.88
3.005	-0.1175	-0.7074	-2.89
3.010	-0.1180	-0.7075	-2.89
3.015	-0.1183	-0.7074	-2.89
3.020	-0.1184	-0.7073	-2.89
3.025	-0.1185	-0.7077	-2.88
3.030	-0.1185	-0.7071	-2.89
3.035	-0.1184	-0.7066	-2.90
3.040	-0.1185	-0.7059	-2.90
3.045	-0.1184	-0.7052	-2.91
3.050	-0.1182	-0.7047	-2.92
3.055	-0.1182	-0.7036	-2.93
3.060	-0.1187	-0.7025	-2.94
3.065	-0.1193	-0.7015	-2.96
3.070	-0.1199	-0.7006	-2.97
3.075	-0.1205	-0.6996	-2.98
3.080	-0.1208	-0.6987	-2.99
3.085	-0.1219	-0.6977	-3.00
3.090	-0.1234	-0.6968	-3.00
3.095	-0.1243	-0.6964	-3.01
3.100	-0.1254	-0.6959	-3.01
3.105	-0.1267	-0.6957	-3.01

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.110	-0.1278	-0.6955	-3.01
3.115	-0.1293	-0.6954	-3.01
3.120	-0.1302	-0.6956	-3.00
3.125	-0.1313	-0.6957	-3.00
3.130	-0.1322	-0.6962	-2.99
3.135	-0.1328	-0.6965	-2.99
3.140	-0.1336	-0.6972	-2.98
3.145	-0.1343	-0.6979	-2.97
3.150	-0.1346	-0.6982	-2.96
3.155	-0.1350	-0.6987	-2.95
3.160	-0.1353	-0.6991	-2.95
3.165	-0.1353	-0.6993	-2.95
3.170	-0.1354	-0.6997	-2.94
3.175	-0.1355	-0.7000	-2.94
3.180	-0.1350	-0.6999	-2.94
3.185	-0.1348	-0.6998	-2.94
3.190	-0.1347	-0.6991	-2.95
3.195	-0.1345	-0.6986	-2.96
3.200	-0.1346	-0.6978	-2.97
3.205	-0.1347	-0.6967	-2.98
3.210	-0.1349	-0.6958	-2.99
3.215	-0.1351	-0.6946	-3.00
3.220	-0.1362	-0.6938	-3.01
3.225	-0.1371	-0.6931	-3.02
3.230	-0.1382	-0.6927	-3.02
3.235	-0.1394	-0.6928	-3.02
3.240	-0.1404	-0.6926	-3.02
3.245	-0.1412	-0.6927	-3.01
3.250	-0.1423	-0.6928	-3.01
3.255	-0.1432	-0.6930	-3.00
3.260	-0.1439	-0.6933	-3.00
3.265	-0.1447	-0.6933	-3.00
3.270	-0.1453	-0.6935	-2.99
3.275	-0.1459	-0.6937	-2.99
3.280	-0.1468	-0.6939	-2.98
3.285	-0.1476	-0.6939	-2.98
3.290	-0.1483	-0.6941	-2.98
3.295	-0.1489	-0.6938	-2.98
3.300	-0.1496	-0.6941	-2.97
3.305	-0.1503	-0.6935	-2.98
3.310	-0.1505	-0.6939	-2.97
3.315	-0.1508	-0.6936	-2.98
3.320	-0.1514	-0.6934	-2.98
3.325	-0.1520	-0.6934	-2.98
3.330	-0.1523	-0.6928	-2.98
3.335	-0.1527	-0.6924	-2.99

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.340	-0.1532	-0.6918	-2.99
3.345	-0.1534	-0.6910	-3.00
3.350	-0.1537	-0.6901	-3.01
3.355	-0.1542	-0.6897	-3.02
3.360	-0.1549	-0.6888	-3.02
3.365	-0.1553	-0.6879	-3.03
3.370	-0.1557	-0.6866	-3.05
3.375	-0.1564	-0.6857	-3.06
3.380	-0.1571	-0.6844	-3.07
3.385	-0.1577	-0.6831	-3.09
3.390	-0.1588	-0.6820	-3.10
3.395	-0.1596	-0.6808	-3.11
3.400	-0.1606	-0.6790	-3.13
3.405	-0.1617	-0.6778	-3.14
3.410	-0.1625	-0.6766	-3.15
3.415	-0.1641	-0.6751	-3.16
3.420	-0.1653	-0.6740	-3.17
3.425	-0.1667	-0.6728	-3.18
3.430	-0.1681	-0.6716	-3.19
3.435	-0.1695	-0.6702	-3.21
3.440	-0.1707	-0.6690	-3.22
3.445	-0.1725	-0.6681	-3.22
3.450	-0.1743	-0.6667	-3.23
3.455	-0.1758	-0.6659	-3.24
3.460	-0.1775	-0.6652	-3.24
3.465	-0.1789	-0.6645	-3.25
3.470	-0.1805	-0.6640	-3.25
3.475	-0.1822	-0.6635	-3.25
3.480	-0.1834	-0.6632	-3.25
3.485	-0.1851	-0.6626	-3.25
3.490	-0.1864	-0.6633	-3.24
3.495	-0.1872	-0.6629	-3.24
3.500	-0.1880	-0.6630	-3.23
3.505	-0.1884	-0.6627	-3.24
3.510	-0.1889	-0.6623	-3.24
3.515	-0.1889	-0.6620	-3.24
3.520	-0.1890	-0.6611	-3.25
3.525	-0.1889	-0.6607	-3.26
3.530	-0.1889	-0.6598	-3.27
3.535	-0.1882	-0.6583	-3.29
3.540	-0.1876	-0.6562	-3.32
3.545	-0.1880	-0.6540	-3.34
3.550	-0.1881	-0.6511	-3.38
3.555	-0.1888	-0.6484	-3.41
3.560	-0.1901	-0.6454	-3.44
3.565	-0.1916	-0.6430	-3.47

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
3.570	-0.1942	-0.6401	-3.49
3.575	-0.1965	-0.6369	-3.52
3.580	-0.2004	-0.6357	-3.52
3.585	-0.2039	-0.6346	-3.52
3.590	-0.2073	-0.6350	-3.51
3.595	-0.2103	-0.6360	-3.48
3.600	-0.2125	-0.6378	-3.45
3.605	-0.2138	-0.6395	-3.42
3.610	-0.2138	-0.6413	-3.40
3.615	-0.2133	-0.6425	-3.39
3.620	-0.2124	-0.6432	-3.38
3.625	-0.2120	-0.6437	-3.38
3.630	-0.2100	-0.6434	-3.39
3.635	-0.2087	-0.6423	-3.41
3.640	-0.2081	-0.6409	-3.43
3.645	-0.2078	-0.6390	-3.45
3.650	-0.2077	-0.6376	-3.47
3.655	-0.2080	-0.6360	-3.49
3.660	-0.2082	-0.6345	-3.51
3.665	-0.2089	-0.6329	-3.52
3.670	-0.2092	-0.6315	-3.54
3.675	-0.2097	-0.6297	-3.56
3.680	-0.2102	-0.6280	-3.58
3.685	-0.2112	-0.6264	-3.60
3.690	-0.2122	-0.6244	-3.62
3.695	-0.2136	-0.6227	-3.63
3.700	-0.2152	-0.6207	-3.65
3.705	-0.2172	-0.6190	-3.66
3.710	-0.2200	-0.6171	-3.67
3.715	-0.2229	-0.6162	-3.67
3.720	-0.2267	-0.6152	-3.67
3.725	-0.2301	-0.6152	-3.65
3.730	-0.2347	-0.6154	-3.63
3.735	-0.2378	-0.6172	-3.59
3.740	-0.2409	-0.6197	-3.54
3.745	-0.2422	-0.6224	-3.51
3.750	-0.2431	-0.6259	-3.46
3.755	-0.2426	-0.6288	-3.43
3.760	-0.2409	-0.6317	-3.40
3.765	-0.2383	-0.6332	-3.39
3.770	-0.2348	-0.6333	-3.41
3.775	-0.2315	-0.6325	-3.43
3.780	-0.2295	-0.6295	-3.48
3.785	-0.2282	-0.6254	-3.53
3.790	-0.2290	-0.6218	-3.57
3.795	-0.2310	-0.6189	-3.60

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.800	-0.2332	-0.6166	-3.62
3.805	-0.2359	-0.6149	-3.63
3.810	-0.2386	-0.6140	-3.63
3.815	-0.2410	-0.6132	-3.62
3.820	-0.2434	-0.6131	-3.61
3.825	-0.2458	-0.6131	-3.60
3.830	-0.2477	-0.6136	-3.59
3.835	-0.2484	-0.6169	-3.54
3.840	-0.2496	-0.6164	-3.54
3.845	-0.2505	-0.6163	-3.54
3.850	-0.2516	-0.6161	-3.54
3.855	-0.2525	-0.6156	-3.54
3.860	-0.2534	-0.6154	-3.54
3.865	-0.2548	-0.6148	-3.54
3.870	-0.2557	-0.6141	-3.54
3.875	-0.2569	-0.6138	-3.54
3.880	-0.2580	-0.6133	-3.54
3.885	-0.2593	-0.6128	-3.54
3.890	-0.2608	-0.6123	-3.54
3.895	-0.2619	-0.6117	-3.54
3.900	-0.2635	-0.6112	-3.54
3.905	-0.2649	-0.6103	-3.54
3.910	-0.2665	-0.6098	-3.54
3.915	-0.2677	-0.6090	-3.54
3.920	-0.2693	-0.6086	-3.54
3.925	-0.2709	-0.6080	-3.54
3.930	-0.2722	-0.6073	-3.54
3.935	-0.2736	-0.6066	-3.54
3.940	-0.2752	-0.6059	-3.54
3.945	-0.2764	-0.6050	-3.54
3.950	-0.2782	-0.6043	-3.54
3.955	-0.2798	-0.6033	-3.54
3.960	-0.2811	-0.6025	-3.55
3.965	-0.2826	-0.6017	-3.55
3.970	-0.2842	-0.6006	-3.55
3.975	-0.2857	-0.5995	-3.56
3.980	-0.2870	-0.5987	-3.56
3.985	-0.2884	-0.5976	-3.56
3.990	-0.2900	-0.5967	-3.56
3.995	-0.2917	-0.5955	-3.57
4.000	-0.2933	-0.5946	-3.57
4.005	-0.2947	-0.5935	-3.57
4.010	-0.2961	-0.5923	-3.58
4.015	-0.2977	-0.5911	-3.58
4.020	-0.2989	-0.5901	-3.59
4.025	-0.3005	-0.5891	-3.59

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.030	-0.3019	-0.5880	-3.60
4.035	-0.3033	-0.5869	-3.60
4.040	-0.3048	-0.5857	-3.61
4.045	-0.3062	-0.5845	-3.61
4.050	-0.3075	-0.5836	-3.61
4.055	-0.3089	-0.5821	-3.62
4.060	-0.3101	-0.5812	-3.63
4.065	-0.3114	-0.5800	-3.63
4.070	-0.3126	-0.5788	-3.64
4.075	-0.3137	-0.5780	-3.64
4.080	-0.3149	-0.5770	-3.64
4.085	-0.3161	-0.5760	-3.65
4.090	-0.3173	-0.5749	-3.65
4.095	-0.3186	-0.5741	-3.65
4.100	-0.3196	-0.5732	-3.66
4.105	-0.3209	-0.5723	-3.66
4.110	-0.3221	-0.5715	-3.66
4.115	-0.3233	-0.5707	-3.66
4.120	-0.3246	-0.5700	-3.66
4.125	-0.3261	-0.5691	-3.66
4.130	-0.3275	-0.5683	-3.66
4.135	-0.3291	-0.5672	-3.67
4.140	-0.3302	-0.5664	-3.67
4.145	-0.3313	-0.5646	-3.68
4.150	-0.3324	-0.5638	-3.68
4.155	-0.3332	-0.5625	-3.69
4.160	-0.3340	-0.5614	-3.70
4.165	-0.3345	-0.5603	-3.71
4.170	-0.3347	-0.5595	-3.72
4.175	-0.3350	-0.5587	-3.72
4.180	-0.3353	-0.5582	-3.73
4.185	-0.3351	-0.5576	-3.73
4.190	-0.3356	-0.5571	-3.74
4.195	-0.3357	-0.5568	-3.74
4.200	-0.3359	-0.5566	-3.74
4.205	-0.3361	-0.5565	-3.74
4.210	-0.3360	-0.5562	-3.74
4.215	-0.3365	-0.5560	-3.74
4.220	-0.3366	-0.5561	-3.74
4.225	-0.3368	-0.5557	-3.74
4.230	-0.3366	-0.5559	-3.74
4.235	-0.3367	-0.5559	-3.74
4.240	-0.3368	-0.5562	-3.74
4.245	-0.3366	-0.5561	-3.74
4.250	-0.3367	-0.5563	-3.74
4.255	-0.3366	-0.5562	-3.74

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.260	-0.3365	-0.5567	-3.74
4.265	-0.3362	-0.5569	-3.73
4.270	-0.3364	-0.5571	-3.73
4.275	-0.3362	-0.5572	-3.73
4.280	-0.3361	-0.5575	-3.73
4.285	-0.3360	-0.5576	-3.73
4.290	-0.3359	-0.5580	-3.72
4.295	-0.3358	-0.5584	-3.72
4.300	-0.3357	-0.5585	-3.72
4.305	-0.3356	-0.5588	-3.72
4.310	-0.3355	-0.5591	-3.71
4.315	-0.3355	-0.5594	-3.71
4.320	-0.3355	-0.5596	-3.71
4.325	-0.3353	-0.5600	-3.71
4.330	-0.3351	-0.5605	-3.70
4.335	-0.3352	-0.5609	-3.70
4.340	-0.3353	-0.5613	-3.69
4.345	-0.3352	-0.5619	-3.68
4.350	-0.3351	-0.5624	-3.68
4.355	-0.3347	-0.5632	-3.67
4.360	-0.3346	-0.5639	-3.67
4.365	-0.3343	-0.5645	-3.66
4.370	-0.3340	-0.5652	-3.66
4.375	-0.3335	-0.5658	-3.65
4.380	-0.3330	-0.5665	-3.65
4.385	-0.3326	-0.5672	-3.64
4.390	-0.3319	-0.5677	-3.64
4.395	-0.3316	-0.5682	-3.64
4.400	-0.3310	-0.5686	-3.64
4.405	-0.3303	-0.5692	-3.63
4.410	-0.3299	-0.5694	-3.63
4.415	-0.3293	-0.5695	-3.64
4.420	-0.3290	-0.5697	-3.64
4.425	-0.3283	-0.5700	-3.64
4.430	-0.3281	-0.5701	-3.64
4.435	-0.3278	-0.5701	-3.64
4.440	-0.3276	-0.5702	-3.64
4.445	-0.3276	-0.5703	-3.64
4.450	-0.3275	-0.5704	-3.64
4.455	-0.3274	-0.5709	-3.63
4.460	-0.3271	-0.5712	-3.63
4.465	-0.3265	-0.5715	-3.63
4.470	-0.3262	-0.5724	-3.62
4.475	-0.3259	-0.5731	-3.62
4.480	-0.3253	-0.5741	-3.61
4.485	-0.3250	-0.5754	-3.60

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.490	-0.3247	-0.5766	-3.59
4.495	-0.3242	-0.5775	-3.58
4.500	-0.3238	-0.5784	-3.57
4.505	-0.3232	-0.5792	-3.57
4.510	-0.3228	-0.5798	-3.56
4.515	-0.3221	-0.5799	-3.57
4.520	-0.3213	-0.5801	-3.57
4.525	-0.3210	-0.5797	-3.57
4.530	-0.3204	-0.5796	-3.58
4.535	-0.3200	-0.5790	-3.59
4.540	-0.3202	-0.5786	-3.59
4.545	-0.3203	-0.5783	-3.60
4.550	-0.3205	-0.5777	-3.60
4.555	-0.3204	-0.5772	-3.61
4.560	-0.3203	-0.5769	-3.61
4.565	-0.3207	-0.5764	-3.61
4.570	-0.3208	-0.5758	-3.62
4.575	-0.3209	-0.5753	-3.63
4.580	-0.3209	-0.5745	-3.63
4.585	-0.3210	-0.5735	-3.64
4.590	-0.3216	-0.5731	-3.65
4.595	-0.3216	-0.5724	-3.65
4.600	-0.3221	-0.5716	-3.66
4.605	-0.3224	-0.5707	-3.67
4.610	-0.3229	-0.5701	-3.67
4.615	-0.3231	-0.5696	-3.68
4.620	-0.3234	-0.5689	-3.68
4.625	-0.3234	-0.5682	-3.69
4.630	-0.3240	-0.5674	-3.70
4.635	-0.3242	-0.5667	-3.70
4.640	-0.3244	-0.5660	-3.71
4.645	-0.3246	-0.5652	-3.72
4.650	-0.3246	-0.5646	-3.72
4.655	-0.3247	-0.5639	-3.73
4.660	-0.3249	-0.5633	-3.74
4.665	-0.3251	-0.5630	-3.74
4.670	-0.3249	-0.5621	-3.75
4.675	-0.3246	-0.5616	-3.76
4.680	-0.3246	-0.5612	-3.76
4.685	-0.3243	-0.5605	-3.77
4.690	-0.3242	-0.5599	-3.78
4.695	-0.3240	-0.5596	-3.79
4.700	-0.3237	-0.5590	-3.80
4.705	-0.3235	-0.5583	-3.80
4.710	-0.3239	-0.5575	-3.81
4.715	-0.3239	-0.5568	-3.82

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.720	-0.3242	-0.5559	-3.83
4.725	-0.3247	-0.5551	-3.84
4.730	-0.3249	-0.5540	-3.85
4.735	-0.3255	-0.5531	-3.85
4.740	-0.3264	-0.5518	-3.86
4.745	-0.3270	-0.5505	-3.87
4.750	-0.3280	-0.5497	-3.87
4.755	-0.3290	-0.5483	-3.88
4.760	-0.3298	-0.5473	-3.89
4.765	-0.3302	-0.5467	-3.89
4.770	-0.3308	-0.5459	-3.90
4.775	-0.3310	-0.5450	-3.91
4.780	-0.3312	-0.5447	-3.91
4.785	-0.3309	-0.5440	-3.92
4.790	-0.3308	-0.5431	-3.93
4.795	-0.3304	-0.5426	-3.94
4.800	-0.3297	-0.5420	-3.95
4.805	-0.3291	-0.5411	-3.97
4.810	-0.3287	-0.5403	-3.98
4.815	-0.3281	-0.5395	-3.99
4.820	-0.3276	-0.5393	-4.00
4.825	-0.3270	-0.5390	-4.01
4.830	-0.3265	-0.5383	-4.02
4.835	-0.3262	-0.5380	-4.02
4.840	-0.3260	-0.5379	-4.03
4.845	-0.3262	-0.5376	-4.03
4.850	-0.3261	-0.5376	-4.03
4.855	-0.3262	-0.5380	-4.02
4.860	-0.3261	-0.5376	-4.03
4.865	-0.3265	-0.5379	-4.02
4.870	-0.3268	-0.5382	-4.02
4.875	-0.3269	-0.5385	-4.01
4.880	-0.3273	-0.5389	-4.01
4.885	-0.3274	-0.5390	-4.00
4.890	-0.3281	-0.5395	-3.99
4.895	-0.3283	-0.5400	-3.99
4.900	-0.3286	-0.5402	-3.98
4.905	-0.3289	-0.5403	-3.98
4.910	-0.3293	-0.5406	-3.97
4.915	-0.3297	-0.5409	-3.96
4.920	-0.3301	-0.5413	-3.96
4.925	-0.3306	-0.5414	-3.95
4.930	-0.3310	-0.5417	-3.95
4.935	-0.3314	-0.5416	-3.95
4.940	-0.3319	-0.5419	-3.94
4.945	-0.3322	-0.5421	-3.93

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.950	-0.3328	-0.5423	-3.93
4.955	-0.3332	-0.5424	-3.92
4.960	-0.3335	-0.5422	-3.92
4.965	-0.3339	-0.5425	-3.92
4.970	-0.3343	-0.5426	-3.91
4.975	-0.3345	-0.5428	-3.91
4.980	-0.3350	-0.5427	-3.91
4.985	-0.3352	-0.5429	-3.90
4.990	-0.3356	-0.5430	-3.90
4.995	-0.3357	-0.5431	-3.90
5.000	-0.3361	-0.5433	-3.89
5.005	-0.3361	-0.5436	-3.89
5.010	-0.3366	-0.5436	-3.88
5.015	-0.3366	-0.5438	-3.88
5.020	-0.3371	-0.5437	-3.88
5.025	-0.3371	-0.5438	-3.88
5.030	-0.3375	-0.5434	-3.88
5.035	-0.3372	-0.5436	-3.88
5.040	-0.3373	-0.5428	-3.89
5.045	-0.3370	-0.5431	-3.89
5.050	-0.3366	-0.5425	-3.90
5.055	-0.3365	-0.5428	-3.89
5.060	-0.3356	-0.5422	-3.91
5.065	-0.3357	-0.5424	-3.91
5.070	-0.3351	-0.5425	-3.91
5.075	-0.3356	-0.5426	-3.90
5.080	-0.3350	-0.5425	-3.91
5.085	-0.3358	-0.5425	-3.90
5.090	-0.3357	-0.5424	-3.91
5.095	-0.3364	-0.5420	-3.91
5.100	-0.3367	-0.5420	-3.90
5.105	-0.3374	-0.5413	-3.91
5.110	-0.3379	-0.5414	-3.90
5.115	-0.3387	-0.5409	-3.90
5.120	-0.3391	-0.5406	-3.90
5.125	-0.3397	-0.5402	-3.90
5.130	-0.3404	-0.5397	-3.90
5.135	-0.3413	-0.5391	-3.90
5.140	-0.3417	-0.5386	-3.91
5.145	-0.3422	-0.5380	-3.91
5.150	-0.3432	-0.5371	-3.91
5.155	-0.3441	-0.5363	-3.91
5.160	-0.3448	-0.5354	-3.92
5.165	-0.3458	-0.5343	-3.93
5.170	-0.3464	-0.5332	-3.93
5.175	-0.3476	-0.5322	-3.94

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.180	-0.3484	-0.5309	-3.94
5.185	-0.3494	-0.5292	-3.96
5.190	-0.3501	-0.5279	-3.97
5.195	-0.3510	-0.5266	-3.97
5.200	-0.3519	-0.5251	-3.98
5.205	-0.3526	-0.5236	-4.00
5.210	-0.3536	-0.5223	-4.00
5.215	-0.3544	-0.5206	-4.02
5.220	-0.3551	-0.5189	-4.03
5.225	-0.3559	-0.5175	-4.04
5.230	-0.3569	-0.5156	-4.05
5.235	-0.3576	-0.5139	-4.07
5.240	-0.3585	-0.5123	-4.08
5.245	-0.3590	-0.5108	-4.09
5.250	-0.3600	-0.5092	-4.10
5.255	-0.3609	-0.5074	-4.12
5.260	-0.3619	-0.5056	-4.13
5.265	-0.3626	-0.5042	-4.14
5.270	-0.3637	-0.5026	-4.15
5.275	-0.3645	-0.5011	-4.16
5.280	-0.3654	-0.4995	-4.17
5.285	-0.3662	-0.4981	-4.18
5.290	-0.3671	-0.4967	-4.19
5.295	-0.3680	-0.4952	-4.20
5.300	-0.3687	-0.4939	-4.20
5.305	-0.3696	-0.4930	-4.21
5.310	-0.3704	-0.4914	-4.22
5.315	-0.3712	-0.4900	-4.23
5.320	-0.3719	-0.4887	-4.23
5.325	-0.3726	-0.4875	-4.24
5.330	-0.3731	-0.4863	-4.25
5.335	-0.3738	-0.4851	-4.26
5.340	-0.3743	-0.4838	-4.27
5.345	-0.3747	-0.4825	-4.28
5.350	-0.3756	-0.4812	-4.29
5.355	-0.3762	-0.4798	-4.30
5.360	-0.3768	-0.4785	-4.31
5.365	-0.3774	-0.4768	-4.32
5.370	-0.3781	-0.4754	-4.33
5.375	-0.3791	-0.4742	-4.33
5.380	-0.3798	-0.4728	-4.34
5.385	-0.3809	-0.4714	-4.35
5.390	-0.3817	-0.4701	-4.36
5.395	-0.3827	-0.4690	-4.36
5.400	-0.3837	-0.4680	-4.36
5.405	-0.3844	-0.4671	-4.37

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.410	-0.3854	-0.4661	-4.37
5.415	-0.3865	-0.4651	-4.37
5.420	-0.3874	-0.4642	-4.37
5.425	-0.3885	-0.4630	-4.37
5.430	-0.3891	-0.4622	-4.38
5.435	-0.3899	-0.4613	-4.38
5.440	-0.3908	-0.4605	-4.38
5.445	-0.3915	-0.4598	-4.38
5.450	-0.3924	-0.4594	-4.38
5.455	-0.3930	-0.4587	-4.38
5.460	-0.3936	-0.4581	-4.38
5.465	-0.3943	-0.4571	-4.38
5.470	-0.3948	-0.4565	-4.39
5.475	-0.3956	-0.4556	-4.39
5.480	-0.3963	-0.4550	-4.39
5.485	-0.3970	-0.4542	-4.39
5.490	-0.3979	-0.4536	-4.39
5.495	-0.3989	-0.4530	-4.39
5.500	-0.3998	-0.4525	-4.38
5.505	-0.4007	-0.4521	-4.38
5.510	-0.4016	-0.4516	-4.37
5.515	-0.4026	-0.4516	-4.36
5.520	-0.4034	-0.4513	-4.36
5.525	-0.4041	-0.4514	-4.35
5.530	-0.4052	-0.4516	-4.34
5.535	-0.4057	-0.4516	-4.34
5.540	-0.4065	-0.4516	-4.33
5.545	-0.4072	-0.4517	-4.32
5.550	-0.4078	-0.4518	-4.31
5.555	-0.4083	-0.4518	-4.31
5.560	-0.4090	-0.4517	-4.30
5.565	-0.4099	-0.4518	-4.29
5.570	-0.4105	-0.4519	-4.29
5.575	-0.4113	-0.4519	-4.28
5.580	-0.4119	-0.4518	-4.27
5.585	-0.4129	-0.4519	-4.26
5.590	-0.4136	-0.4519	-4.26
5.595	-0.4146	-0.4520	-4.25
5.600	-0.4155	-0.4520	-4.24
5.605	-0.4165	-0.4521	-4.23
5.610	-0.4173	-0.4525	-4.22
5.615	-0.4182	-0.4528	-4.20
5.620	-0.4191	-0.4531	-4.19
5.625	-0.4200	-0.4535	-4.18
5.630	-0.4206	-0.4539	-4.17
5.635	-0.4212	-0.4542	-4.16

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.640	-0.4217	-0.4543	-4.15
5.645	-0.4223	-0.4548	-4.14
5.650	-0.4228	-0.4551	-4.14
5.655	-0.4232	-0.4553	-4.13
5.660	-0.4237	-0.4556	-4.12
5.665	-0.4243	-0.4556	-4.12
5.670	-0.4247	-0.4557	-4.11
5.675	-0.4252	-0.4559	-4.10
5.680	-0.4257	-0.4563	-4.10
5.685	-0.4259	-0.4564	-4.09
5.690	-0.4263	-0.4560	-4.09
5.695	-0.4266	-0.4563	-4.09
5.700	-0.4269	-0.4561	-4.09
5.705	-0.4273	-0.4561	-4.08
5.710	-0.4277	-0.4565	-4.07
5.715	-0.4276	-0.4562	-4.08
5.720	-0.4277	-0.4562	-4.08
5.725	-0.4280	-0.4560	-4.08
5.730	-0.4282	-0.4562	-4.07
5.735	-0.4285	-0.4562	-4.07
5.740	-0.4288	-0.4559	-4.07
5.745	-0.4285	-0.4553	-4.08
5.750	-0.4288	-0.4551	-4.08
5.755	-0.4289	-0.4550	-4.08
5.760	-0.4290	-0.4545	-4.08
5.765	-0.4289	-0.4541	-4.09
5.770	-0.4290	-0.4536	-4.09
5.775	-0.4292	-0.4532	-4.09
5.780	-0.4293	-0.4525	-4.10
5.785	-0.4296	-0.4519	-4.10
5.790	-0.4294	-0.4511	-4.11
5.795	-0.4293	-0.4504	-4.12
5.800	-0.4293	-0.4497	-4.13
5.805	-0.4292	-0.4487	-4.14
5.810	-0.4292	-0.4479	-4.15
5.815	-0.4291	-0.4472	-4.15
5.820	-0.4292	-0.4462	-4.16
5.825	-0.4290	-0.4449	-4.18
5.830	-0.4290	-0.4441	-4.19
5.835	-0.4291	-0.4432	-4.20
5.840	-0.4292	-0.4421	-4.21
5.845	-0.4293	-0.4411	-4.22
5.850	-0.4293	-0.4402	-4.22
5.855	-0.4295	-0.4390	-4.23
5.860	-0.4297	-0.4378	-4.24
5.865	-0.4300	-0.4368	-4.25

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.870	-0.4301	-0.4357	-4.26
5.875	-0.4306	-0.4347	-4.27
5.880	-0.4308	-0.4336	-4.28
5.885	-0.4309	-0.4327	-4.28
5.890	-0.4314	-0.4317	-4.29
5.895	-0.4316	-0.4309	-4.29
5.900	-0.4320	-0.4300	-4.30
5.905	-0.4324	-0.4293	-4.30
5.910	-0.4324	-0.4286	-4.31
5.915	-0.4326	-0.4280	-4.31
5.920	-0.4325	-0.4271	-4.32
5.925	-0.4326	-0.4265	-4.33
5.930	-0.4326	-0.4256	-4.34
5.935	-0.4324	-0.4247	-4.35
5.940	-0.4328	-0.4238	-4.35
5.945	-0.4331	-0.4229	-4.36
5.950	-0.4330	-0.4218	-4.37
5.955	-0.4335	-0.4209	-4.38
5.960	-0.4338	-0.4200	-4.38
5.965	-0.4343	-0.4190	-4.39
5.970	-0.4348	-0.4184	-4.39
5.975	-0.4353	-0.4178	-4.39
5.980	-0.4364	-0.4174	-4.38
5.985	-0.4367	-0.4169	-4.38
5.990	-0.4372	-0.4172	-4.38
5.995	-0.4381	-0.4169	-4.37
6.000	-0.4385	-0.4172	-4.36

2 Measurement Quality

Reliable band: from 45 MHz (S11 error σ = 1.0e-03); the points below it are kept in the plot (shaded, outside the scale) and in the tables, but are not reliable.
Spike removal: on (isolated point, 2nd difference > 6 x local noise and > S11 error budget, not shared with another sweep); replaced 1 point(s): Short at 4885 MHz ($|\Delta S11|$ = 1.4e-03).
Outliers shared by several sweeps, left alone (instrument artefacts cancelled by the calibration): 3835 MHz.

- Unreliable below 45 MHz: 8 point(s) (3 mm probe, S11 error σ = 1.0e-03). There the reference liquids resolve ε_r only to ±17.2 at 5 MHz … ±2.1 at 40 MHz per σ (usable: $\leq$ ±2). Values kept, but left out of the chart scale and of the summary medians; start the sweep at $\geq$ 45 MHz.

- Gn shows no radiation signature even at the top of the sweep (3 GHz – 6 GHz): mean arg(Gn) = -108° (radiation: -90°, median deviation 20°) and $|Gn| \propto f^{0.4}$ (radiation: f^3). Over the whole sweep it holds noise and the inconsistency between the reference liquids instead (its term is ≈7% of the Ref 1 (Distilled Water) admittance), so the full method's radiation correction is empirical here: compare with the simplified cross-check, especially for samples whose ε_r lies outside the range of the reference liquids.

- 2 frequency point(s) excluded: reference liquids nearly indistinguishable ($|S11_ref1 - S11_ref2| < 1e\text{-}02$). Consider starting the sweep above the affected range.